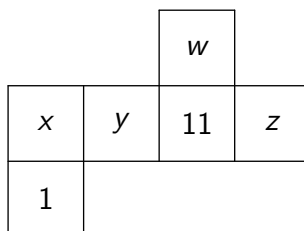




1. Let $f(x)$ be the function $f(x) = 3x + 10$. For what value of x does $f(f(x)) = x$?
2. The difference between a positive fraction and its reciprocal is $\frac{9}{20}$. Find the sum of the fraction and its reciprocal.
3. The diagram shows the net of a cube. On each face there is an integer: 1, 11, w , x , y , and z .



If each of the numbers w , x , y , and z equals the average of the numbers written on the four faces of the cube adjacent to it, find the value of x .

4. Solve $(x^2 - 3x + 1)^2 - 3(x^2 - 3x + 1) + 2 = 0$
5. Consider the following system of seven linear equations with seven unknowns:

$$\begin{aligned}a + b + c + d + e &= 1, \\b + c + d + e + f &= 2, \\c + d + e + f + g &= 3, \\d + e + f + g + a &= 4, \\e + f + g + a + b &= 5, \\f + g + a + b + c &= 6, \\g + a + b + c + d &= 7.\end{aligned}$$

What is g ?

6. Let a and b be real numbers such that

$$\begin{aligned}\frac{1}{a^2} + \frac{3}{b^2} &= 2018a, \\ \frac{3}{a^2} + \frac{1}{b^2} &= 290b.\end{aligned}$$

Find $\frac{ab}{b-a}$.



1. The ratio of the number of tables to chairs in a restaurant is $4 : 9$. If I remove 2 tables and 4 chairs, the ratio becomes $7 : 16$. How many tables and chairs are left?
2. Starting at 1 PM, Bernie goes for a walk at a leisurely pace of 2 miles per hour. At 2 PM, Joe starts jogging at the rate of 5 miles per hour, in the same direction as Bernie, and starting at the same point. At what time does Joe catch up to Bernie?
3. Cameron had two candles, one of which was 32 cm longer than the other. She lit the taller candle at 3 p.m. and lit the shorter candle at 7 p.m. At 9 p.m., they were both the same height. The taller one was completely burned out at 10 p.m. and the shorter one was completely burned at midnight. The two candles burned at different, but constant, rates. What were the original heights of the two candles?
4. Supergran runs from her chalet to the top of the mountain. She knows that if she runs at a speed of 6 mph she will arrive at 1 pm, whereas if she leaves at the same time and runs at 10 mph, she will arrive at 11 am. At what speed should she walk if she wants to arrive at 12 noon?
5. David drives from his home to the airport to catch a flight. He drives 35 miles in the first hour, but realizes that he will be 1 hour late if he continues at this speed. He increases his speed by 15 miles per hour for the rest of the way to the airport and arrives 30 minutes early. How many miles is the airport from his home?
6. Angela and Barry share a piece of land. The ratio of the area of Angela's portion to the area of Barry's portion is $3 : 2$. They each grow corn and peas on their portion of land. The entire piece of land is covered by corn and peas in the ratio $7 : 3$. On Angela's portion of the land, the ratio of corn to peas is $4 : 1$. What is the ratio of corn to peas for Barry's portion?
7. The grass in a field grows at a constant rate. If 240 sheep were to graze on it, then the field would become bare in 5 weeks. If 200 sheep were to graze on it, then the field would become bare in 7 weeks. If 170 sheep were to graze, how many weeks would it take before the field becomes bare?



AMC10/12 Camp Class 1 Handout 3: Team Contest

You have 50 minutes to complete all 6 problems on this contest. You can earn a maximum of 50 points.

1. **The 'Team' Challenge.** Find five of the following things. Be prepared to show off any physical items to the rest of the class (5 points, 1 point each):
 - Something that everyone on your team loves to do during the summer
 - A non-American food that everyone on your team loves
 - A snack that everyone on your team has at home
 - A non-math topic that the people on your team could give a 30-minute presentation on without any preparation
 - A book that everyone on your team has physical copies of at home
 - A piece of technology at least 15 years old that everyone on your team has at home
 - A song everyone on your team knows all the words to
 - A movie or TV show that everyone on your team dislikes
 - An interesting thing that everyone on your team has in common
 - A common thing no one on your team has ever done



2. **KenKen.** Fill in this puzzle so that:

- Each of the 49 small squares contains a digit between 1 and 7 inclusive.
- No number is repeated in a row or column.
- The numbers in each heavily outlined set of squares, called cages, must combine (in some order) to produce the target number in the top corner using the mathematical operation indicated.

Cages with just one square should be filled in with the target number in the top corner. Also, note that a number can be repeated within a cage as long as it is not in the same row or column. (7 points)

6−	1−		24×			2−
	840×		2÷	7	9+	
60×				252×		
	3+	2−				3−
		210×	6−		12+	
3÷						13+
	5+		30×			

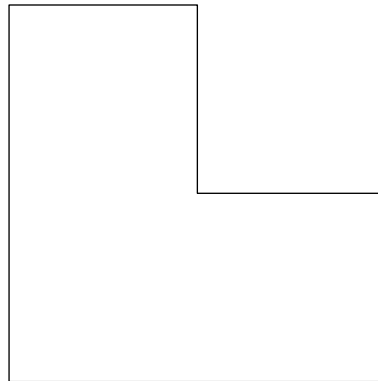
3. **The Logician's Society.** Many logicians meet once a year for the annual Logician's Society meeting. Logicians either always tell the truth or always lie. Two logicians are said to be "of the same type" if they are both truth-tellers or both liars. Answer the following questions about some of the peculiar events of this annual meeting. (8 points)

- (a) 100 logicians go out for dinner. Each logician announces, "You are all liars." How many of these logicians are liars? (2 points)
- (b) Five logicians sit around a circular table. Each one announces, "I am seated next to at least one liar." How many of these logicians are liars? (2 points)
- (c) One of the biggest events at the Logician's Society meeting is a live taping of the game show *Do You Know Logic?*, where a non-logician enlists some logicians' help to win a prize. In one round, there are three boxes labeled X, Y, and Z, and exactly one of them contains a prize. Five logicians make the following statements:

- A: Box X contains the prize.
- B: Box Y contains the prize.
- C: A and B are not both liars.
- D: Either A is a liar, or B is a truth-teller, or both.
- E: Either I am a liar, or C and D are of the same type, or both.

If the contestant wants to win the prize, which box should they pick? (4 points)

4. **Geometric Division Puzzles.** These questions deal with the following figure, formed by putting together three congruent squares: (8 points)



- (a) Divide the above figure into four congruent figures, each of which is similar to the original figure. (2 points)
- (b) Divide the above figure into eight congruent quadrilaterals. (2 points)
- (c) Divide the above figure into nine congruent figures, each of which is similar to the original figure. (4 points)

5. **Which is Greater?** In each of the following problems, you will be given two quantities A and B . Determine whether $A > B$, $A = B$, or $A < B$. (14 points)

(a) $A = 2^{1005}3^{1000} + 2^{1000}3^{1005}$, $B = 2^{1004}3^{1001} + 2^{1003}3^{1002} + 2^{1002}3^{1003} + 2^{1001}3^{1004}$ (2 points)

(b) $A = \underbrace{3939 \dots 39}_{1234 \text{ digits}} \times \underbrace{4141 \dots 41}_{1234 \text{ digits}}$, $B = \underbrace{3838 \dots 38}_{1234 \text{ digits}} \times \underbrace{4242 \dots 42}_{1234 \text{ digits}}$ (2 points)

(c) $A = \sqrt[8]{8}$, $B = \sqrt[9]{9}$ (3 points)

(d) $A = 31^{19}$, $B = 17^{24}$ (3 points)

(e) $A = 50^{99}$, $B = 99!$ (4 points)

6. **Mathematical Games.** Each of the following mathematical games is played by two players, Ash and Blair, who alternate turns with Ash going first. In a game, a player has a *winning strategy* if, regardless of what the other player does, there are moves that they can make which guarantee that they will win. Answer the following questions about mathematical games and their winning strategies (8 points):

- (a) This game begins with a pile of n toothpicks, where n is a positive integer. On each player's turn, they must remove 1, 3, or 4 toothpicks from the pile. The player who removes the last toothpick wins the game.
- Who has a winning strategy when $n = 20$? (1 point)
 - For how many values of n such that $1 \leq n \leq 1000$ does Blair have a winning strategy? (3 points)
- (b) This game begins with two piles of m and n stones, where m and n are positive integers. On each player's turn, they can take as many stones as they want from a single pile, but must take at least one stone. The player who takes the last stone wins.
- Who has a winning strategy when $m = 16$ and $n = 17$? (1 point)
 - For how many pairs of integers (m, n) with $1 \leq m \leq 100$ and $1 \leq n \leq 100$ does Blair have a winning strategy? (3 points)



1 Systems of Equations

The two most common techniques for solving systems of equations are **substitution** and **elimination**.

Example. Solve the system

$$\begin{aligned}2x + y &= 3, \\ -4x + 3y &= 1.\end{aligned}$$

Solution. We can multiply the first equation by 2, which gives us

$$\begin{aligned}4x + 2y &= 6, \\ -4x + 3y &= 1.\end{aligned}$$

We can then add the equations, which eliminates x , and gives us $5y = 7$. Then $y = \frac{7}{5}$. Substituting this value into the first equation, we get

$$2x + \frac{7}{5} = 3.$$

Then $2x = \frac{8}{5}$, so $x = \frac{4}{5}$. Therefore, the solution is $(x, y) = (\frac{4}{5}, \frac{7}{5})$.

Alternatively, we could have solved for y in the first equation to get $y = 3 - 2x$. We can then substitute this expression for y into the second equation, which gives us a linear equation in x . ■

Example. Suppose that $4^a = 5$, $5^b = 6$, $6^c = 7$, and $7^d = 8$. What is $a \cdot b \cdot c \cdot d$? (2005 AMC 10B, #17)

Solution. The first equation states $4^a = 5$. Since we have a term of 5^b in the second equation, we can raise both sides to the power of b , which gives us

$$4^{ab} = 5^b.$$

Hence, $4^{ab} = 6$. (In a sense, we have substituted $5 = 4^a$ into the second equation.)

Since we have a term of 6^c in the third equation, we can raise both sides to the power of c , which gives us

$$4^{abc} = 6^c.$$

Hence, $4^{abc} = 7$.

Finally, raising both sides to the power of d , we get

$$4^{abcd} = 7^d.$$

Hence, $4^{abcd} = 8$.

Since $4 = 2^2$ and $8 = 2^3$, we can write $2^{2abcd} = 2^3$. Therefore, $2abcd = 3$, so $abcd = \frac{3}{2}$. ■

Exercise. If x , y , and z are positive with $xy = 24$, $xz = 48$, and $yz = 72$, then find $x + y + z$. (2001 AMC 10, #10)

Exercise. In the magic square shown, the sums of the numbers in each row, column, and diagonal are the same. Five of these numbers are represented by v , w , x , y , and z . Find $y + z$. (2001 AMC 10, #22)

v	24	w
18	x	y
25	z	21

2 Rate Problems

For rate problems, the fundamental relationship is given by distance = rate $\times$ time.

Example. A train is 280 m long. It takes 60 seconds to clear a tunnel (from the time it enters to the time it leaves), and 20 seconds to pass a telephone pole. Find the length of the tunnel.

Solution. Since the train is 280 m long, and it takes 20 seconds to pass a telephone pole, the speed of the train is

$$\frac{280 \text{ m}}{20 \text{ sec}} = 14 \text{ m/sec.}$$

Let d be the length of the tunnel in meters. Then to clear the tunnel, the train travels $d + 280$ m. Therefore,

$$d + 280 = 14 \cdot 60,$$

which implies $d = 560$. ■

Example. It takes Clea 60 seconds to walk down an escalator when it is not operating, and only 24 seconds to walk down the escalator when it is operating. How many seconds does it take Clea to ride down the operating escalator when she just stands on it? (2012 AMC 10B, #13)

Solution. Let c be Clea's walking speed, let e be the speed of the escalator, and let d be the length of the escalator.

It takes Clea 60 seconds to walk down the escalator when it is not moving, so

$$d = 60c.$$

When the escalator is moving, Clea takes only 24 seconds. Also, Clea's speed on the moving escalator is $c + e$, so

$$d = 24(c + e).$$

Then $60c = 24c + 24e$, which simplifies to $e = \frac{3}{2}c$.

When Clea rides down the escalator, it takes $\frac{d}{e}$ seconds. Since $d = 60c$ and $e = \frac{3}{2}c$,

$$\frac{d}{e} = \frac{60c}{\frac{3}{2}c} = 40.$$

■

Exercise. Yan is somewhere between his home and the stadium. To get to the stadium he can walk directly to the stadium, or else he can walk home and then ride his bicycle to the stadium. He rides 7 times as fast as he walks, and both choices require the same amount of time. What is the ratio of Yan's distance from his home to his distance from the stadium? (2007 AMC 10A, #13)

Exercise. Andrea and Lauren are 20 kilometers apart. They bike toward one another with Andrea traveling three times as fast as Lauren, and the distance between them decreasing at a rate of 1 kilometer per minute. After 5 minutes, Andrea stops biking because of a flat tire and waits for Lauren. After how many minutes from the time they started to bike does Lauren reach Andrea? (2009 AMC 10A, #20)



A **sequence** is a list of numbers. Each number in a sequence is called a **term**. Sequences can be **finite** or **infinite** depending on whether they have a finite or infinite number of terms. When we add the terms of a sequence, we form a **series**.

There are two special types of sequences and series which commonly show up on contests: Arithmetic sequences and series, and geometric sequences and series.

1 Arithmetic Sequences and Series

Definition 1.

- An **arithmetic sequence** is a sequence where the difference between two consecutive terms is always the same. For a given arithmetic sequence, this difference is known as the **common difference**.
- When we add the terms of a finite arithmetic sequence, we get an **arithmetic series**.

Suppose an arithmetic sequence has first term a and common difference d . To go from the first term to the n th term of an arithmetic sequence, we start with a and add the common difference $(n - 1)$ times, so the n th term will be $a + (n - 1)d$. To find the formula for the sum of an arithmetic series, consider the example below:

Example 2. Evaluate the sum $1 + 2 + 3 + \cdots + 99 + 100$.

Solution. Let the given sum be S . This is an arithmetic series with a first term of 1, a common difference of 1, and 100 terms. To find this sum, we can reverse the order of terms in this sum and add it to our original sum. In doing so, we pair the 1st and 100th terms, 2nd and 99th terms, and so on to get 100 pairs, each of which sums to 101:

$$\begin{array}{r} S = 1 + 2 + 3 + \cdots + 100 \\ S = 100 + 99 + 98 + \cdots + 1 \\ \hline 2S = 101 + 101 + 101 + \cdots + 101 \end{array}$$

We have $2S = 100 \cdot 101$, so $S = \frac{100 \cdot 101}{2} = \boxed{5050}$.

More generally, to sum any arithmetic series, one can pair the 1st and n th terms, 2nd and $(n - 1)$ st terms, and so on to get n pairs, each of which has a sum equal to the sum of the first and n th terms. Each term of the series has been counted twice here, so it follows that an arithmetic series with first term a , common difference d , and n terms has sum $n \cdot \frac{2a + (n - 1)d}{2}$, or (Number of terms) $\cdot \frac{\text{First term} + \text{Last term}}{2}$.

2 Geometric Sequences and Series

Definition 3.

- A **geometric sequence** is a sequence where the ratio between two consecutive terms is always the same. For a given geometric sequence, this difference is known as the **common ratio**.
- When we add the terms of a geometric sequence, we get a **geometric series**.

Suppose a geometric sequence has first term a and common ratio r . To go from the first term to the n th term of a geometric sequence, we start with a and multiply by the common ratio $(n - 1)$ times, so the n th term will be $a \cdot r^{n-1}$. To find the formula for the sum of a geometric series, consider the example below:

Example 4. Evaluate the sum $4^0 + 4^1 + 4^2 + 4^3 + \cdots + 4^{100}$.



Solution. Let the given sum be S . This is a geometric series with a first term of $4^0 = 1$, a common ratio of 4, and 101 terms. To find this sum, we can multiply every term of this sum by the common ratio and subtract it from our original sum, and in doing so, most of our terms cancel.

$$\begin{array}{rcl} S & = & 4^0 + 4^1 + 4^2 + \cdots + 4^{100} \\ -(4S & = & 4^1 + 4^2 + \cdots + 4^{100} + 4^{101}) \\ \hline -3S & = & 4^0 + 0 + 0 + \cdots + 0 - 4^{101} \end{array}$$

Thus, we have $S = \boxed{\frac{4^{101} - 1}{3}}$.

More generally, to sum any finite geometric series, we can multiply the entire sum by the common ratio and subtract it from the original series. Every term will cancel except for the first term of the original series and the last term of the second series, so it follows that a finite geometric series with first term a , common ratio r , and n terms has sum $\frac{a(r^n - 1)}{r - 1}$.

Notice that each term of our geometric series above was bigger than the last, so our series wouldn't have a finite sum if we extended it infinitely. On the other hand, it turns out that if the common ratio r of a geometric series satisfies $|r| < 1$ (or $-1 < r < 1$), an infinite geometric series with common ratio r will actually have a finite sum. To find this sum, we can use the same idea as above:

Example 5. Evaluate the sum $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \cdots \dots + \left(-\frac{1}{3}\right)^k + \cdots$.

Solution. Let the given sum be S . This is an infinite geometric series with a first term of 1 and a common ratio of $-\frac{1}{3}$. To find this sum, we can multiply the entire sum by the common ratio and subtract it from the original sum, which will make every term but the first cancel.

$$\begin{array}{rcl} S & = & 1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \frac{1}{81} + \cdots \\ -\left(-\frac{S}{3} & = & -\frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \frac{1}{81} + \cdots\right) \\ \hline \frac{4S}{3} & = & 1 + 0 + 0 + 0 + 0 + \cdots \end{array}$$

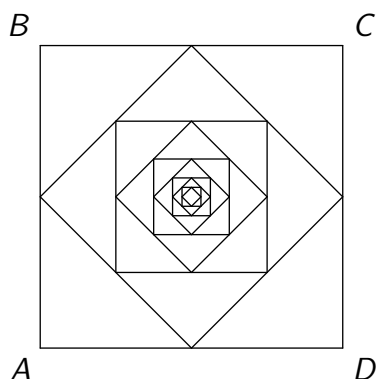
Thus, we have $S = \boxed{\frac{3}{4}}$.

We can use this same technique to evaluate the sum of any infinite geometric series with first term a and common ratio r , so long as $|r| < 1$. (Again, this sum is not defined when $|r| \geq 1$.) All terms but the first term of the original series will cancel, and we find that this sum is equal to $\frac{a}{1 - r}$.

For more about sequences and series, consult Chapter 21 of our *Introduction to Algebra* textbook, Chapter 10 of our *Intermediate Algebra* textbook, and Chapter 24 of *The Art of Problem Solving: Volume 1*.



1. Paul starts at 1 and counts by threes: $1, 4, 7, 10, \dots$. At the same time and at the same speed, Penny counts backwards from 2017 by fives: $2017, 2012, 2007, 2002, \dots$. Find the one number that both Paul and Penny count at the same time.
2. Mary divides a circle into 12 sectors. The central angles of these sectors, measured in degrees, are all integers and they form an arithmetic sequence. What is the degree measure of the smallest possible sector angle?
3. Square $ABCD$ has side length 10. A second square is drawn by connecting the midpoints of the sides of $ABCD$. This process is continued infinitely, as shown in the picture below. What is the sum of the perimeters of all the squares?



4. Suppose that $a_1, a_2, a_3, \dots$ is an arithmetic sequence with $a_1 + a_2 + \dots + a_{100} = 100$ and $a_{101} + a_{102} + \dots + a_{200} = 200$. What is the value of $a_2 - a_1$?
5. How many distinct geometric sequences of integers are there with first term 1 and last term 2^{24} ?
6. A sequence consists of 2010 terms. Each term after the first is 1 larger than the previous term. The sum of the 2010 terms is 5307. When every second term is added up, starting with the first term and ending with the second-to-last term, what is the sum?
7. What is the sum of all the unit fractions with terminating decimal expansions? (A unit fraction is a fraction of the form $\frac{1}{n}$, where n is a positive integer.)



1. Let $a_1, a_2, a_3, \dots$ be an infinite geometric sequence of positive numbers. If this sequence has the property that $a_n = 2a_{n+1} + 3a_{n+2}$, what is the common ratio?
2. A sequence of positive real numbers $\{a_1, a_2, a_3, \dots\}$ has the property that for $i \geq 2$, each a_i is equal to the sum of all the previous terms. If $a_{19} = 99$, then what is a_{20} ?
3. The sum of an infinite geometric series is a positive number S , and the second term in the series is 1. What is the smallest possible value of S ?
4. For each positive integer k , let S_k denote the increasing arithmetic sequence of integers whose first term is 1 and whose common difference is k . For example, S_3 is the sequence $1, 4, 7, 10, \dots$. For how many values of k does S_k contain the term 2005?
5. An infinite geometric series has sum 2020. A new series, obtained by squaring each term of the original series, has 10 times the sum of the original series. What is the common ratio of the original series?
6. Let $a_1 = 1 + \sqrt{2}$ and for each $n \geq 1$ define $a_{n+1} = 2 - \frac{1}{a_n}$. What is the product $a_1 a_2 a_3 \cdots a_{200}$?
7. Let the sequence $\{a_n\}_{n \geq 0}$ be defined by $a_0 = \frac{1}{2}$ and $a_n = 1 + (a_{n-1} - 1)^2$ for $n \geq 1$. Find the infinite product $a_0 \cdot a_1 \cdot a_2 \cdot a_3 \cdots$.

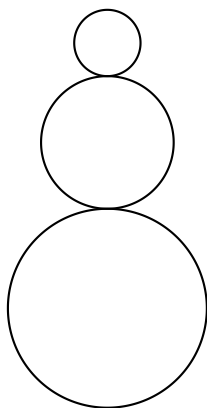


1. If $u = -6$ and $x = \frac{1}{3}(3 - 4u)$, then what is the value of x ?
(A) -23 (B) -7 (C) 2 (D) 9 (E) 25

1.

2. A snowman is built by stacking three spheres with their centers aligned vertically. The spheres have radii of 10 cm, 20 cm and 30 cm. How tall is the snowman?

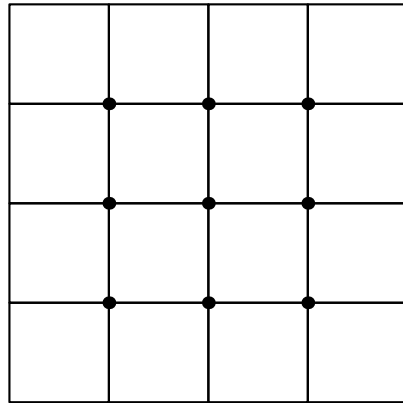
2.



- (A) 90 cm (B) 100 cm (C) 110 cm
(D) 120 cm (E) 130 cm



3. The nine interior intersection points on a 4 by 4 grid of squares are shown. How many interior intersection points are there on a 12 by 12 grid of squares?



(A) 100 (B) 121 (C) 132 (D) 144 (E) 169

3.

4. How many sets of two prime numbers $\{p, q\}$ have the property that $p + q = 100$?

(A) 5 (B) 6 (C) 7 (D) 10 (E) 12

4.

5. The odd numbers from 5 to 21 are used to build a 3 by 3 magic square. (In a magic square, the numbers in each row, the numbers in each column, and the numbers on each diagonal have the same sum.) If 5, 9 and 17 are placed as shown, what is the value of x ?

	5	
9		17
x		

(A) 7 (B) 11 (C) 13 (D) 15 (E) 19

5.

6. How many distinct positive integers can be expressed in the form $ABCD - DCBA$, where $ABCD$ and $DCBA$ are 4-digit positive integers? (Here A, B, C , and D are digits, possibly equal, with $A \neq 0$ and $D \neq 0$.)

(A) 108 (B) 161 (C) 162 (D) 322 (E) 323

6.

7. How many subsets of $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ have the property that no two of its elements differ by more than 5? For example, count the sets $\{3\}$, $\{2, 5, 7\}$, and $\{5, 6, 7, 8, 9\}$ but not the set $\{1, 3, 5, 7\}$.

(A) 160 (B) 192 (C) 256 (D) 384 (E) 2048

7.

8. How many real numbers x satisfy the equation

$$(3^x)^{x+2} + (4^x)^{x+2} - (6^x)^{x+2} = 1?$$

(A) 1 (B) 2 (C) 3 (D) 4 (E) 8

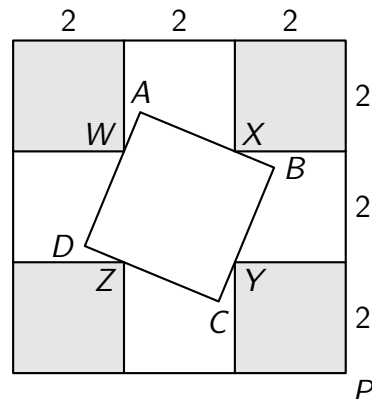
8.

9. Each of the 48 faces of eight $1 \times 1 \times 1$ cubes is randomly painted either blue or green. The probability that these eight cubes can then be assembled into a $2 \times 2 \times 2$ cube in a way so that its surface is solid green can be written $\frac{p^m}{q^n}$, where p and q are prime numbers and m and n are positive integers. Find $p + q + m + n$.

(A) 49 (B) 77 (C) 95 (D) 107 (E) none of these

9.

10. In the diagram, four squares of side length 2 are placed in the corners of a square of side length 6. Each of the points W , X , Y , and Z is a vertex of one of the small squares, and the point P is a vertex of the large square. Square $ABCD$ can be constructed with sides passing through W , X , Y , and Z . The maximum possible distance from A to P is closest to which number?



(A) 5 (B) 5.25 (C) 5.5 (D) 5.75 (E) 6

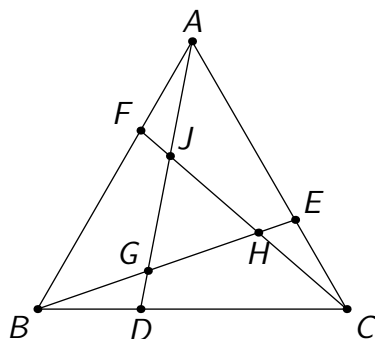
10.

11. The number 104,060,465 is divisible by a five-digit prime number. What is the sum of the digits of this prime number?
 (A) 11 (B) 13 (C) 14 (D) 16 (E) 17

11.

12. On equilateral $\triangle ABC$, point D lies on $\overline{BC}$ a distance 1 from B , point E lies on $\overline{AC}$ a distance 1 from C , and point F lies on $\overline{AB}$ a distance 1 from A . Segments $\overline{AD}$, $\overline{BE}$, $\overline{CF}$ intersect in pairs at points G , H , and J which are the vertices of another equilateral triangle. The area of $\triangle ABC$ is twice the area of $\triangle GHJ$. The side length of $\triangle ABC$ can be written as $\frac{r + \sqrt{s}}{t}$, where r , s , and t are relatively prime positive integers. Find $r + s + t$.

12.



- (A) 28 (B) 29 (C) 30 (D) 31 (E) 32



1 Quadratics

Definition 1. A *quadratic function* is a function of the form $q(x) = ax^2 + bx + c$, where a, b, c are real numbers and $a \neq 0$. Any solution to the equation $q(x) = 0$ is called a **root** of the quadratic.

We won't be talking too much about quadratics in this class, but here are a few things to know about quadratics that are useful to know on contests. In a quadratic equation $ax^2 + bx + c = 0$, the number $b^2 - 4ac$ is known as the **discriminant** of the quadratic. Then:

- When the discriminant is positive, then $q(x)$ has two real solutions;
- When the discriminant is zero, then $q(x)$ has one real solution (i.e. a *double root*);
- When the discriminant is negative, then $q(x)$ has no real solutions;
- When the discriminant is a perfect square, then $q(x)$ has rational solutions.

Also, a quadratic $q(x) = ax^2 + bx + c$ achieves its minimum or maximum value at the vertex of the quadratic, which has x -coordinate $-\frac{b}{2a}$. This can be seen by completing the square:

$$\begin{aligned} ax^2 + bx + c &= a\left(x^2 + \frac{b}{a}x\right) + c \\ &= a\left(\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2}\right) + c \\ &= a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}. \end{aligned}$$

Since $\left(x + \frac{b}{2a}\right)^2$ will always be non-negative, it follows that if a is positive, then the quadratic can be minimized at $x = -\frac{b}{2a}$ (by letting $\left(x + \frac{b}{2a}\right)^2$ be as small as possible), and similarly, if a is negative, then the quadratic can be maximized at $x = -\frac{b}{2a}$.

Finally, we can find the sum and product of the roots of a quadratic. Suppose the quadratic $ax^2 + bx + c$ has roots at $x = r$ and $x = s$. Then we have that $ax^2 + bx + c = a(x - r)(x - s)$. Expanding the right-hand side, we have that $ax^2 + bx + c = ax^2 - a(r + s)x + ars$. Since these two quadratics are equal, their coefficients must be equal, so it follows that $r + s = -\frac{b}{a}$ and $rs = \frac{c}{a}$.

2 Polynomials

Definition 2. A *polynomial* is a function of the form $p(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_2x^2 + a_1x + a_0$, where $a_n, a_{n-1}, \dots, a_1, a_0$ are real numbers and $a_n \neq 0$. The a_i are called the **coefficients** of $p(x)$, and n is the **degree** of $p(x)$. If $a_n = 1$, the polynomial is **monic**. Any solution to the equation $p(x) = 0$ is known as a **root** of the polynomial.

Example 3. $f(x)$ is a degree 2 polynomial satisfying $f(0) = 7$, $f(1) = 4$, and $f(2) = 5$. Find $f(x)$.

Solution. Since $f(x)$ is a degree 2 polynomial, it must be of the form $f(x) = ax^2 + bx + c$. By substituting $x = 0, 1, 2$ respectively, we find that $c = 7$, $a + b + c = 4$, and $4a + 2b + c = 5$. Solving the system, we find that $a = 2$ and $b = -5$, so $f(x) = 2x^2 - 5x + 7$.

We notice a few interesting things in this example: Firstly, for a polynomial $p(x)$, we have that $p(0)$ is the constant term and $p(1)$ is the sum of the coefficients of p . Also, a degree 2 polynomial is just a quadratic function. (A degree 3 polynomial is known as a **cubic**, and a degree 4 polynomial is known as a **quartic**.)



Polynomials can be added, subtracted or multiplied in the usual way (by combining like terms or distributing). One can show that any polynomial $p(x)$ of degree n has exactly n complex roots (counting with multiplicity), and can be written in the form $p(x) = a_n(x - r_1)(x - r_2) \cdots (x - r_n)$, where the r_i are the roots of the polynomial. Equivalently, if $p(x)$ has a root r , then $p(x)$ can be written $p(x) = (x - r)q(x)$, where $q(x)$ is a polynomial of degree $n - 1$. This is sometimes known as the **Factor Theorem**.

Example 4. A cubic polynomial $p(x)$ has roots at $x = 1, 2, 3$. If $p(4) = 1$, what is $p(5)$?

Solution. Since $p(x)$ is cubic and has roots at $x = 1, 2, 3$, we know $p(x)$ has no other roots, so we can write $p(x) = c(x - 1)(x - 2)(x - 3)$ for some leading coefficient c . Substituting $x = 4$, we find that $p(4) = c(4 - 1)(4 - 2)(4 - 3) = 6c = 1$, so $c = \frac{1}{6}$. So $p(5) = \frac{1}{6}(5 - 1)(5 - 2)(5 - 3) = \frac{1}{6} \cdot 4 \cdot 3 \cdot 2 = \boxed{4}$.

A simple but powerful technique when working with polynomials is to **compare coefficients**: If two polynomials are equal, then their coefficients must also be equal.

Example 5. A quadratic function $q(x)$ satisfies $2q(x) - q(x + 1) = x^2$ for every real number x . What is $q(x)$?

Solution. Suppose that $q(x) = ax^2 + bx + c$. Then we have that

$$\begin{aligned} 2q(x) - q(x + 1) &= 2(ax^2 + bx + c) - (a(x + 1)^2 + b(x + 1) + c) \\ &= ax^2 + (-2a + b)x + (-a - b + c). \end{aligned}$$

Since this must be equal to x^2 , we can compare coefficients to see that $a = 1$, $-2a + b = 0$, $-a - b + c = 0$. Solving, we find that $a = 1$, $b = 2$, $c = 3$, so $\boxed{q(x) = x^2 + 2x + 3}$.

Notice that we implicitly used the idea of comparing coefficients above to find the sum and product of the roots of a quadratic equation, and we can generalize our conclusions to all polynomials: If we write $p(x) = a_n(x - r_1)(x - r_2) \cdots (x - r_n) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_2x^2 + a_1x + a_0$, expand, and compare the coefficients of the resulting expressions, we find that

$$\begin{aligned} r_1 + r_2 + r_3 + \cdots + r_n &= (-1)^1 \frac{a_{n-1}}{a_n}, \\ r_1 r_2 + r_1 r_3 + r_1 r_4 + \cdots + r_{n-1} r_n &= (-1)^2 \frac{a_{n-2}}{a_n}, \\ r_1 r_2 r_3 + r_1 r_2 r_4 + \cdots + r_{n-2} r_{n-1} r_n &= (-1)^3 \frac{a_{n-3}}{a_n}, \\ &\vdots \\ r_1 r_2 r_3 \cdots r_n &= (-1)^n \frac{a_0}{a_n}, \end{aligned}$$

where the sums on the left-hand side of each equation are found by taking the sum of all the roots, the sum of all products of two distinct roots, the sum of all products of three distinct roots, and so on. These are known as **Vieta's formulas**. (If these look scary to you, make sure you understand what these formulas mean for $n = 2, 3, 4$; these are the cases that will most commonly arise on contests.)

Example 6. Let r, s, t be the roots of the polynomial $x^3 + 2x^2 + 3x + 4$. Find $\frac{1}{r} + \frac{1}{s} + \frac{1}{t}$.

Solution. By clearing denominators, we see that $\frac{1}{r} + \frac{1}{s} + \frac{1}{t} = \frac{rs+st+rt}{rst}$. Vieta's formulas tell us that $rs + st + rt = 3$ and $rst = -4$, so $\frac{1}{r} + \frac{1}{s} + \frac{1}{t} = \boxed{-\frac{3}{4}}$.

For more about polynomials, consult Chapters 6-8 of our *Intermediate Algebra* textbook and Chapter 6 of *The Art of Problem Solving: Volume 2*.



AMC10/12 Camp Class 3 Handout 1: Vieta's Formulas

1. Find the sum of the squares of the values x that satisfy $\frac{1}{x} + \frac{2}{x+3} + \frac{3}{x+6} = 1$.
2. For real numbers a, b , and c the polynomial $p(x) = 3x^7 - 291x^6 + ax^5 + bx^4 + cx^2 + 134x - 2$ has 7 real roots whose sum is 97. Find the sum of the reciprocals of those 7 roots.
3. Let a, b , and c be three distinct one-digit numbers. What is the maximum value of the sum of the roots of the equation $(x-a)(x-b) + (x-b)(x-c) = 0$?
4. The roots of $x^3 - 7x^2 + cx + 27$ can be arranged to form a geometric sequence. What is the value of c ?
5. Let $f(x) = x^4 - x^3 - 7x^2 + 2x - 5$. Suppose that $g(x)$ is a monic polynomial whose roots are each 3 greater than those of $f(x)$. What is the coefficient of the x^2 term of $g(x)$?
6. The polynomial $x^3 - ax^2 + bx - 2010$ has three positive integer roots. What is the smallest possible value of a ?
7. Suppose that a, b, c are the roots of $f(x) = x^3 - 5x^2 + 12x + 1$. Find $\frac{a^2 + b^2}{c^2} + \frac{a^2 + c^2}{b^2} + \frac{b^2 + c^2}{a^2}$.



1. For some constant k , the polynomial $p(x) = 3x^2 + kx + 117$ has the property that $p(1) = p(10)$. Find $p(20)$.
2. If $f(x)$ is a monic degree 5 polynomial with $f(1) = 1$, $f(2) = 3$, $f(3) = 6$, $f(4) = 10$, and $f(5) = 15$, find $f(6)$.
3. Three of the roots of $x^4 + ax^2 + bx + c$ are 2, -3 , and 5. What is the value of $a + b + c$?
4. A quartic polynomial P has four roots: $\frac{1}{4}$, $\frac{1}{2}$, 2, 4. If $P(1) = 1$, what is $P(0)$?
5. The cubic polynomials $p(x)$ and $q(x)$ satisfy $p(1) = q(2)$, $p(3) = q(4)$, $p(5) = q(6)$, $p(7) = q(8) + 13$. Find $p(9) - q(10)$.
6. Let P be a polynomial satisfying $P(x + 1) + P(x - 1) = x^3$ for all real numbers x . Find the value of $P(12)$.
7. Let p , q , and r be the distinct roots of the polynomial $x^3 - 22x^2 + 80x - 67$. It is given that there exist real numbers A , B , and C such that

$$\frac{1}{s^3 - 22s^2 + 80s - 67} = \frac{A}{s - p} + \frac{B}{s - q} + \frac{C}{s - r}$$

for all $s \notin \{p, q, r\}$. What is $\frac{1}{A} + \frac{1}{B} + \frac{1}{C}$?





In this handout, we provide a brief introduction to some fundamental counting techniques. For more, consult Chapters 1-4 of our *Introduction to Counting and Probability* textbook and Chapter 25 of *The Art of Problem Solving: Volume 1*.

1 Counting as Arithmetic

We begin with an example:

Example 1. *There are 7 puppies and 9 kittens at the animal shelter.*

(a) *If I want to select only one animal to take home with me, how many possibilities are there for which animal I can take?*

(b) *If I want to select one puppy and one kitten to take home with me, how many possibilities are there for which two animals I can take?*

Solution.

(a) We can think of this as doing one of two tasks: Selecting a puppy to take home, or selecting a kitten to take home. There are 7 ways to do the first task, 9 ways to do the second task, and these tasks cannot occur at the same time. Therefore, there are $7 + 9 = \boxed{16}$ ways to take an animal home with me.

(b) We can think of the overall procedure of taking home two animals as being broken down into two stages: Selecting a puppy to take home, and then selecting a kitten to take home. For each of the 7 possible choices of a puppy I can take home, there are 9 possible choices of a kitten I can take home, and which puppy I take home doesn't affect which kitten I take home. Therefore, there are $7 \cdot 9 = \boxed{63}$ ways to take two animals home with me.

Many counting problems are grounded in two fundamental counting principles demonstrated by the above example:

The Addition Principle: If a first task can be done in n ways and a second task in m ways, and if these tasks cannot occur at the same time, then there are $n + m$ ways to do either task.

The Multiplication Principle: If a procedure can be broken into two stages, and if there are n outcomes in the first stage and m outcomes in the second stage (independent of the choice in the first stage), then the total procedure has $n \cdot m$ composite outcomes.

Note that the Addition and Multiplication Principles can be generalized to more than two tasks and more than two stages of a procedure, respectively.

Example 2. *How many different ways are there to arrange four different books on a shelf?*

Solution. We can place the books from left to right. There are 4 ways to choose the one book to place on the leftmost end of the shelf. No matter which book we choose, there are 3 ways to choose one of the remaining books to place to the immediate right of the first book. There are then 2 ways to choose a remaining book to place to the immediate right of the second, and we're forced to place the remaining book on the right end of the shelf. Therefore, there are $4 \cdot 3 \cdot 2 \cdot 1 = \boxed{24}$ ways to arrange the books on a shelf.

Notice that in the previous example, we're still able to apply the Multiplication Principle as even though our choice of which books go on the shelf are not independent, the *number* of choices at each stage remains the same. Also, the product seen at the end of the problem shows up frequently enough in counting problems that we have a name for it:

Definition 3. The quantity n **factorial** is denoted by $n!$ and equal to $n \cdot (n - 1) \cdot (n - 2) \cdots 2 \cdot 1$.

By convention, we define $0! = 1$. (One way to think about this is that there's only one way to arrange zero books on a shelf - do nothing!)

2 Basic Counting Techniques

In this section, we discuss four basic techniques that can be applied to a wide variety of counting problems.

2.1 Constructive Counting

In many counting problems, thinking about how we could construct one example of something that we want to count gives rise to a way to count everything we want. This general idea is called **constructive counting**.

Example 4. How many three-digit positive integers only have even digits?

Solution. We can construct a three-digit positive integer with only even digits by choosing a hundreds digit that is one of 2, 4, 6, or 8, and then tens and ones digits that are one of 0, 2, 4, 6, and 8. There are 4 ways to choose the hundreds digit and 5 ways to choose each of the tens and ones digits, so there are $4 \cdot 5 \cdot 5 = \boxed{100}$ such integers.

2.2 Casework

Many counting problems can be solved by dividing what we're trying to count into several categories, where the items in each category are easier to count. These categories are called **cases**, and the overall strategy is called **casework**. The key to solving problems by casework is to make sure you don't miss any cases!

Example 5. How many three-digit positive integers have at least one digit which is a 5?

Solution. We consider three different cases of three-digit positive integers: Those where exactly one digit is a 5, those where exactly two digits are 5's, and those where all three digits are 5's.

Case 1: One digit is a 5. We can break this into subcases based on which of the three digits is a 5.

- **Subcase 1a: The hundreds digit is a 5.** In this case, there are 9 ways to pick each of the tens and units digits, since these digits cannot be 5's. Therefore, there are $9 \cdot 9 = 81$ possibilities for this subcase.
- **Subcase 1b: The tens digit is a 5.** In this case, there are 8 ways to pick the hundreds digit and 9 ways to pick the units digits so they are not 5's, so there are $8 \cdot 9 = 72$ possibilities for this subcase.
- **Subcase 1c: The units digit is a 5.** In a similar vein to subcase 1b, there are 8 ways to pick the hundreds digit and 9 ways to pick the tens digits so they are not 5's, so there are $8 \cdot 9 = 72$ possibilities for this subcase.

Adding the possibilities for all the subcases, there are $81 + 72 + 72 = 225$ three-digit positive integers where exactly one digit is a 5.

Case 2: Two digits are 5's. We can break this into three subcases based on which of the three digits is *not* a 5. If the hundreds digit is not a 5, there are 8 ways to choose its value. If the tens or the ones digit is not

a 5, there are 9 ways to choose its value. So there are $8 + 9 + 9 = 26$ three-digit positive integers where exactly two digits are 5's.

Case 3: Three digits are 5's. There is only one such positive integer, 555.

Adding the results from all cases up, there are $225 + 26 + 1 = \boxed{252}$ three-digit positive integers with at least one digit which is a 5.

2.3 Complementary Counting

Sometimes it's useful to consider what we *don't* want to count instead of what we *do* want to count. This general technique is called **complementary counting**.

Example 6. *How many three-digit positive integers have at least one digit which is a 5?*

Solution. We can construct a three-digit positive integer by choosing one of 9 possible hundreds digits, 10 possible tens digits, and 10 possible units digits, so we see that there are $9 \cdot 10 \cdot 10 = 900$ three-digit positive integers. On the other hand, we can also construct a three-digit positive integer that does *not* contain a 5 by choosing each digit so it is not equal to 5. We see that such an integer must have one of 8 possible hundreds digits, 9 possible tens digits, and 9 possible units digits, so there are $8 \cdot 9 \cdot 9 = 648$ three-digit positive integers that do not contain a 5. Therefore, it follows there are $900 - 648 = \boxed{252}$ three-digit positive integers which have at least one digit that is a 5.

This solution is a lot quicker than the one we saw earlier! While casework is sometimes the only useful way to approach a problem, **if you find yourself in complicated casework, it's useful to at least consider complementary counting.**

2.4 Counting with Restrictions

Often, counting problems will include restrictions - conditions that influence what we want to count. When dealing with restrictions, it is often useful to **consider the restriction first**.

Example 7. *How many ways are there to seat a group of 7 friends in 7 chairs if Blaine and Blair both insist on sitting next to Adrian?*

Solution. The most difficult person to seat is Adrian, since they must have free seats to both their left and right, so we seat them first. Adrian cannot sit in either of the end chairs (since they would not have a free seat in one direction), so they can sit in one of the 5 middle chairs. Then, there are 2 ways for Blaine and Blair to sit in the chairs immediately next to Adrian - either Blaine can sit to Adrian's left and Blair can sit to Adrian's right, or vice versa. Finally, there are $4!$ ways to seat the remaining 4 people in the 4 remaining chairs from left to right. So there are a total of $5 \cdot 2 \cdot 4! = \boxed{240}$ ways to seat these 7 friends.

3 Correcting for Overcounting

So far, the techniques we've looked at have all involved counting everything we want to count once and only once. But it can often be useful to consider counting some or all of the objects we want to count *more* than once (or **overcounting**), and then correcting for our overcount.

3.1 Correcting by Subtracting

When we deliberately count *some* (but not *all*) of the objects we want to count more than once, it is useful to keep in mind which objects we've overcounted and exactly how many times we've counted them.

Example 8. How many three-digit positive integers either start with a 5 or end with a 5 (or both)?

Solution. A three-digit positive integer that starts with a 5 has 10 possible tens digits and 10 possible units digits, so there are $10 \cdot 10 = 100$ of these integers. A three-digit positive integer that ends with a 5 has 9 possible hundreds digits and 10 possible tens digits, so there are $9 \cdot 10 = 90$ of these integers.

At this stage, we're not quite done - we've counted all the integers we want to count, but we've counted the integers which both start and end with a 5 twice (once in each case). To ensure they're only counted once, we can subtract off the number of such integers once. We can pick 10 possible tens digits for such a number, so there are 10 of these integers. Therefore, there are $100 + 90 - 10 = \boxed{180}$ three-digit positive integers that either start or end with a 5.

3.2 Correcting by Dividing

On the other hand, we can deliberately count *all* of the objects we want to count the same number of times each, but more than once. Then, to find the number of objects we want to count, we can divide by the number of times we've counted each object.

Example 9. I have 9 different books. How many ways are there to arrange three of them on a shelf?

The Multiplication Principle tells us that the answer to this is $9 \cdot 8 \cdot 7 = 504$, but we consider a different solution here.

Solution. Equivalently, consider the number of possibilities for the leftmost three books when we arrange all nine of our books on a shelf. We know there are $9!$ ways to arrange all nine of our books on a shelf. Then, for each possible arrangement of the leftmost three books on our shelf, there are $6!$ ways to arrange the other six books to their right. All of these are counted in the quantity $9!$, but we want to only count each arrangement of the leftmost three books once, so we divide by $6!$. So the total number of arrangements is $\frac{9!}{6!} = \boxed{504}$.

We can use a similar argument to find the number of ways to arrange k of n books on a shelf. This quantity comes up frequently enough that we have a name for it:

Definition 10. The number of *permutations* of k objects from a group of n objects is the number of ways to choose k objects with respect to order from a group of n objects. This is denoted $P(n, k)$, and is equal to

$$P(n, k) = n \cdot (n - 1) \cdot (n - 2) \cdots (n - k + 1) = \frac{n!}{(n - k)!}.$$

All of the examples we've seen so far have involved choosing objects in a way where order matters. We can also use this strategy to figure out how to choose objects if order *doesn't* matter:

Example 11. I have 9 different books. How many ways can I choose three of them to donate?

Solution. We know there are $9 \cdot 8 \cdot 7$ ways to choose three of our books with respect to order. However, this overcounts, since we want to count the number of ways to choose three books *without* respect to order. To correct for overcounting, notice that $9 \cdot 8 \cdot 7$ counts each possible unordered choice of three books $3! = 6$ times, once for each way the three books can be ordered. Therefore, the number of ways to choose three books to donate is $\frac{9 \cdot 8 \cdot 7}{3!} = \boxed{84}$.

Again, we can use a similar argument to derive the number of ways to choose k of n books, where order doesn't matter. This quantity also comes up frequently enough that we have a name for it:

Definition 12. The number of **combinations** of k objects from a group of n objects is the number of ways to choose k objects without respect to order from a group of n objects. This is denoted as $C(n, k)$ or $\binom{n}{k}$, is pronounced "n choose k", and is equal to

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n \cdot (n-1) \cdot (n-2) \cdots (n-k+1)}{k!}.$$

The strategy of dividing to correct for overcounting can be used in a number of other problems, too:

Example 13. How many ways are there to rearrange the letters in the word MISSISSIPPI?

Solution. Pretend for a moment that every letter is different, so that our word is really $MI_1S_1S_2I_2S_3S_4I_3P_1P_2I_4$. There are $11!$ ways to arrange the letters in this word. This counts every arrangement of MISSISSIPPI many times: For each arrangement of MISSISSIPPI, we can choose an arrangement of $MI_1S_1S_2I_2S_3S_4I_3P_1P_2I_4$ by picking one of $4!$ ways to order the I's, one of $4!$ ways to order the S's, and one of $2!$ ways to order the P's. Each of these choices is made independently, so it follows that

we've counted each arrangement of MISSISSIPPI $4! \cdot 4! \cdot 2!$ times. Therefore, there are

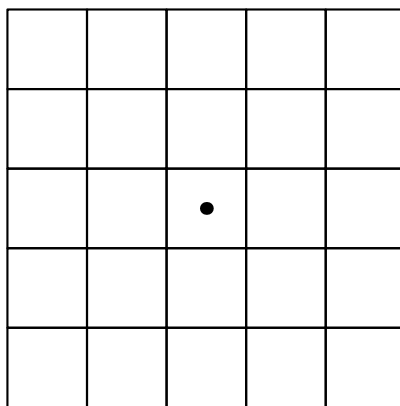
$\frac{11!}{4! \cdot 4! \cdot 2!}$

arrangements of MISSISSIPPI.

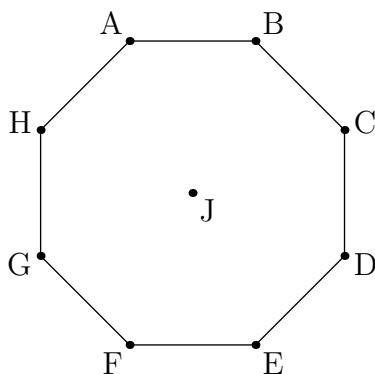


AMC10/12 Camp Class 4 Handout 1: Counting Problems

1. How many even integers are there between 200 and 700 whose digits are all different and come from the set $\{1, 2, 5, 7, 8, 9\}$?
2. How many integers n are there with the property that the product of the digits of n is 0, where $5000 \leq n \leq 6000$?
3. Find the number of squares where the sides of the square are segments in the following diagram such that the dot is inside the square.

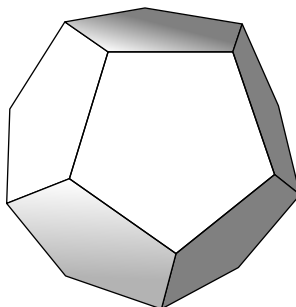


4. The regular octagon $ABCDEFGH$ has its center at J . Each of the vertices and the center are to be associated with one of the digits 1 through 9, with each digit used once, in such a way that the sums of the numbers on the lines AJE , BJF , CJG , and DJH are all equal. In how many ways can this be done?



5. How many triangles with positive area have all their vertices at points (i, j) in the coordinate plane, where i and j are integers between 1 and 5, inclusive?
6. There are 10 people standing equally spaced around a circle. Each person knows exactly 3 of the other 9 people: The 2 people standing next to them, as well as the person directly across the circle. How many ways are there for the 10 people to split up into 5 pairs so that the members of each pair know each other?
7. As shown in the figure below, a regular dodecahedron (the polyhedron consisting of 12 congruent regular pentagonal faces) floats in space with two horizontal faces. Note that there is a ring of five

slanted faces adjacent to the top face, and a ring of five slanted faces adjacent to the bottom face. How many ways are there to move from the top face to the bottom face via a sequence of adjacent faces so that each face is visited at most once and moves are not permitted from the bottom ring to the top ring?





1. Find the number of rearrangements of the word MATHMEET that begin and end with the same letter, such as TAMEMHET.
2. Hal and Janice have a book report due tomorrow. To save time, they decide that Hal will read every third page of the book (pages 3, 6, 9, ...) and Janice, a faster reader, will read the even-numbered pages. If the book has 100 pages, how many pages do they miss altogether?
3. Let (a, b, c, d) be an ordered quadruple of not necessarily distinct integers, each one of them in the set $\{0, 1, 2, 3\}$. For how many such quadruples is it true that $a \cdot d - b \cdot c$ is odd? (For example, $(0, 3, 1, 1)$ is one such quadruple, because $0 \cdot 1 - 3 \cdot 1 = -3$ is odd.)
4. Each vertex of convex pentagon $ABCDE$ is to be assigned a color. There are 6 colors to choose from, and the ends of each diagonal must have different colors. How many different colorings are possible?
5. At a party, seven people contribute a gift, and are randomly given one of the seven contributed gifts (which may be their original gift). A cycle is a set of n people whose gifts ended up going in a circle, for some ordering of people in the set: The first person's gift ended up with the second person, whose gift ended up with the third person, and so on until the n th person's gift ended up with the first person. It is possible to have a cycle with one person, where they got their own gift back, or a cycle with two people, where they swapped gifts.

Observe that every person will always be in exactly one cycle. How many ways can the seven people be given gifts so there are exactly two cycles among the seven people?

6. Amy, Beth, and Carol listen to four different songs and discuss which ones they like. No song is liked by all three. Furthermore, for each of the three pairs of the girls, there is at least one song liked by those two girls but disliked by the third. In how many different ways is this possible?
7. Adam, Benin, Chiang, Deshawn, Esther, and Fiona have internet accounts. Some, but not all, of them are internet friends with each other, and none of them has an internet friend outside this group. Each of them has the same number of internet friends. In how many different ways can this happen?

(You can assume that internet friendship is symmetric: In other words, if person x is internet friends with person y , then person y is also internet friends with person x .)



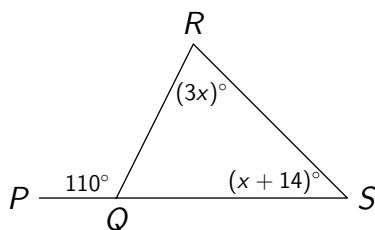
AMC10/12 Camp
Class 4 Handout 3: Mini-AMC Contest

1. Which of the following fractions has the greatest value?

(A) $\frac{44444}{55555}$ (B) $\frac{5555}{6666}$ (C) $\frac{666}{777}$ (D) $\frac{77}{88}$ (E) $\frac{8}{9}$

1.

2. In the diagram, PQS is a straight line. What is the value of x ?



2.

- (A) 19 (B) 21.5 (C) 24 (D) 32 (E) 62

3. In the addition problem shown, X , Y , and Z represent distinct digits. What is the value of $X + Y + Z$?

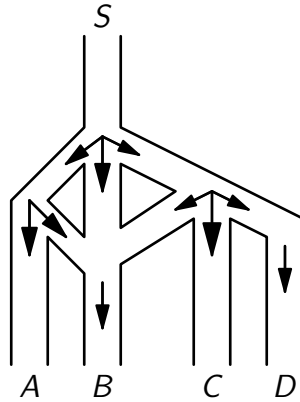
$$\begin{array}{r} X \ Y \ Z \\ X \ Y \ Z \\ + \quad \quad Y \ Z \\ \hline 1 \ 6 \ 7 \ 5 \end{array}$$

3.

- (A) 8 (B) 10 (C) 15 (D) 20 (E) 22

4. Harry the Hamster is put in a maze, and he starts at point S . The paths are such that Harry can move forward only in the direction of the arrows. At any junction, he is equally likely to choose any of the forward paths. What is the probability that Harry ends up at B ?

4.



- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{11}{18}$ (D) $\frac{13}{18}$ (E) $\frac{2}{3}$

5. For real numbers a and b , define the operation $a\#b = ab - a - b$. Which of the following is true?
 I. For all real numbers x, y , $x\#y = y\#x$.
 II. For all real numbers x, y, z , $(x\#y)\#z = x\#(y\#z)$.
 III. There exists a real number d such that for all real numbers x , $d\#x = x$.

5.

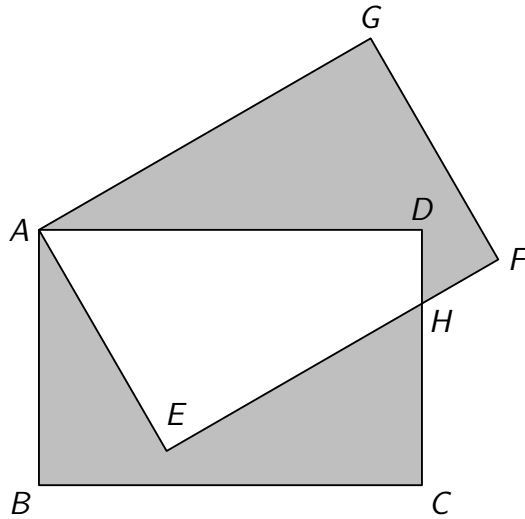
- (A) I only (B) II only (C) I and II only
 (D) I and III only (E) I, II, and III

6. If x and y are integers with $(y - 1)^{x+y} = 4^3$, then the number of possible values for x is

6.

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 8

7. In the diagram, $ABCD$ is a rectangle with $AB = 12$ and $BC = 18$. Rectangle $AEFG$ is formed by rotating $ABCD$ about A through an angle of 30° . The total area of the shaded regions can be expressed as $a - b\sqrt{c}$, where a, b , and c are positive integers, and c is not divisible by the square of any prime. Find $a + b + c$.



- (A) 519 (B) 531 (C) 535 (D) 555 (E) 567

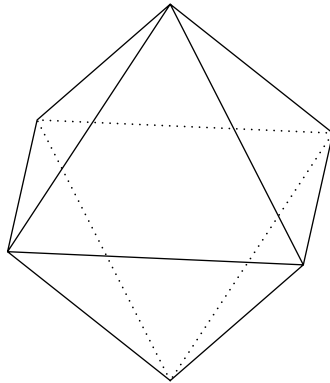
8. Suppose that $S(x) = 2x^3 - 9x^2 + L$, and the three real roots of $S(x)$ form an arithmetic sequence. If L can be expressed in the form $\frac{p}{q}$, where p and q are relatively prime positive integers, what is $p + q$?

- (A) 19 (B) 23 (C) 29 (D) 31 (E) 37

9. What is the remainder when $3^{5^{35}}$ is divided by 35?

- (A) 3 (B) 6 (C) 8 (D) 17 (E) 33

10. An ant begins at a vertex of an octahedron. The ant moves along one edge at a time. Each time the ant reaches a vertex, it randomly chooses to next walk along any of the edges extending from that vertex (including the edge it just arrived from). The probability that after walking along exactly five (not necessarily distinct) edges, the ant finds itself at its starting vertex can be expressed as $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $p + q$.



(A) 19 (B) 21 (C) 27 (D) 37 (E) 75

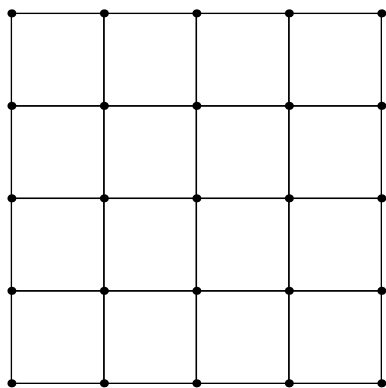
11. The polynomial $P(x) = a_0 + a_1x + a_2x^2 + \cdots + a_8x^8 + 1337x^9$ has the property that $P(\frac{1}{k}) = \frac{1}{k}$ for every integer k where $1 \leq k \leq 9$. If $P(\frac{1}{10})$ can be expressed in the form $\frac{p}{q}$, where p and q are relatively prime positive integers, what are the last three digits of p ?

(A) 000 (B) 336 (C) 337 (D) 663 (E) 664

10.

11.

12. 25 dots are arranged in a 5×5 grid, as shown:



12.

Define the distance between any two dots in this grid as the minimum number of horizontal and vertical steps along the grid lines it takes to get from one dot to the other. For example, two adjacent dots are a distance 1 apart, and two dots at opposite corners of the grid are a distance of 8 apart. What is the mean distance between two distinct dots in the grid?

- (A) $\frac{20}{7}$ (B) $\frac{72}{25}$ (C) $\frac{35}{12}$ (D) 3 (E) $\frac{10}{3}$



In this handout, we provide a brief introduction to some fundamental probability techniques. For more, consult Chapters 7-10 of our *Introduction to Counting and Probability* textbook.

1 Probability as Counting

Definition 1. The *probability* of an event occurring is the proportion of equally likely outcomes which correspond to our event. Notation-wise, let $P(A)$ be the probability of an event A occurring.

The above definition suggests that we can approach many probability problems as two counting problems: First, we figure out how many *equally likely* outcomes there are, then count how many of these outcomes are “successful”. Our probability is the ratio of the number of successful outcomes to the total number of possible outcomes. For example, we can say the probability a fair coin flip turns up heads is $\frac{1}{2}$ since there are two equally likely outcomes - heads and tails - and only flipping heads is “successful”.

Example 2. A bag contains 10 tiles numbered with the positive integers 1 through 10. Winnie removes two distinct tiles from the bag at random. What is the probability that both of Winnie’s tiles have prime numbers on them?

There’s two different ways to approach this problem, depending on what we consider the equally likely outcomes to be: Choosing tiles with regard to order, or choosing tiles without regard to order.

Solution 1. Consider an “outcome” to be any choice of two tiles Winnie can draw in order. There are $10 \cdot 9$ ways to draw two tiles: 10 ways to draw the first tile, and 9 ways to draw the second tile afterwards. There are $4 \cdot 3$ ways to draw two prime-numbered tiles: There are 4 ways Winnie’s first tile can be prime-numbered, since the only prime numbers between 1 and 10 are 2, 3, 5, and 7. If Winnie’s first tile is prime-numbered, she can draw a second prime-numbered tile in 3 ways. So the probability is $\frac{4 \cdot 3}{10 \cdot 9} = \frac{2}{15}$.

Solution 2. Consider an “outcome” to be any choice of two tiles Winnie can draw, where order doesn’t matter. There are $\binom{10}{2} = 45$ ways to draw two tiles without regard to order. Then, there are $\binom{4}{2} = 6$ ways Winnie can choose two prime-numbered tiles without regard to order. So the probability is $\frac{6}{45} = \frac{2}{15}$.

Both solutions gave us the same answer! That being said, we need to ensure that we determine the number of successful outcomes and the possible outcomes in a consistent way, so that each successful outcome corresponds to exactly one possible outcome. If we don’t, we might end up with the following incorrect solution:

Bogus Solution. There are $\binom{10}{2} = 45$ ways to draw two tiles from the bag. There are $4 \cdot 3$ ways for the tiles I draw to be prime-numbered: 4 ways for the first tile, and 3 ways for the second tile afterwards. So the probability is $\frac{4 \cdot 3}{45} = \frac{4}{15}$.

In this case, each of our “possible outcomes” actually corresponded to *two* of our “successful outcomes”, since we can draw the same set of two tiles in two different orders. So our answer is twice what it should’ve been!

2 Probability as Arithmetic

Many counting techniques have analogues in the probability world. For example, if A and B are two events (which need not be mutually exclusive), we have $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. This is

analogous to counting all possible outcomes for A and B individually, and subtracting the double-counted outcomes to correct for overcounting.

Example 3. A bag contains 20 tiles: 10 red tiles numbered with the positive integers 1 through 10, and 10 green tiles numbered with the positive integers 1 through 10. If I remove a tile from the bag at random, what is the probability it is prime-numbered or green (or both)?

Solution. The probability of choosing a prime-numbered tile is $\frac{8}{20} = \frac{2}{5}$, and the probability of choosing a green tile is $\frac{10}{20} = \frac{1}{2}$. However, adding these probabilities double-counts outcomes where we draw a prime-numbered green tile, so we need to subtract the probability of choosing a prime-numbered green tile, or $\frac{4}{20} = \frac{1}{5}$. So the desired probability is $\frac{2}{5} + \frac{1}{2} - \frac{1}{5} = \boxed{\frac{7}{10}}$.

Notice that we get the same probability as if we'd counted outcomes - there are $8 + 10 - 4 = 14$ tiles which are either prime-numbered or green, and 20 possible tiles we could have drawn. In fact, if we didn't simplify any of our fractions, our final probability computation would've read $\frac{8}{20} + \frac{10}{20} - \frac{4}{20}$, which makes this a lot easier to see! When working with probabilities, it's often useful to keep fractions unsimplified until the very end of a problem.

Just as we use the Multiplication Principle when counting outcomes for independent processes, we can also multiply probabilities of independent events: We call two or more events *independent* if the outcome of one event can't affect the outcome of any other events. If A and B are two independent events, then $P(A \text{ and } B) = P(A) \times P(B)$. To see why this is true, consider that if A and B are independent, we have that

$$\begin{aligned} P(A) \times P(B) &= \frac{\text{Successful outcomes for } A}{\text{Total outcomes for } A} \times \frac{\text{Successful outcomes for } B}{\text{Total outcomes for } B} \\ &= \frac{\text{Successful outcomes for } A \text{ and } B}{\text{Total outcomes for } A \text{ and } B} \\ &= P(A \text{ and } B). \end{aligned}$$

Example 4. Six fair coins are flipped. What is the probability that none of them come up heads?

Solution. The probability of any one coin not coming up heads is $\frac{1}{2}$. Since each coin flip is independent, the probability no coin flip comes up heads is $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \boxed{\frac{1}{64}}$.

Notice that we get the same probability as if we'd counted outcomes - there's only one successful outcome here, and $2^6 = 64$ possible sequences of coin flips.

On the other hand, when our events aren't independent, we need to be a bit more careful. Consider our example from the previous section:

Example 5. A bag contains 10 tiles numbered with the positive integers 1 through 10. I remove two distinct tiles from the bag at random. What is the probability both tiles I remove have prime numbers on them?

Solution 3. In order for both tiles I remove to be prime-numbered, I first need to remove a prime-numbered tile, and then remove a second prime-numbered tile **after** I draw the first prime-numbered tile. The probability the first tile I remove is prime-numbered is $\frac{4}{10} = \frac{2}{5}$, and the probability the second tile I draw **after** the first prime-numbered tile is $\frac{3}{9} = \frac{1}{3}$. So the desired probability is $\frac{2}{5} \cdot \frac{1}{3} = \boxed{\frac{2}{15}}$.

Just as we can complementary count, we can also find complementary probabilities: Sometimes it's easier to calculate the probability of an event not occurring than it is to calculate the probability of an event occurring. Then, we have

$$P(\text{A doesn't occur}) = 1 - P(\text{A occurs}).$$

Example 6. Six fair coins are flipped. What is the probability at least one of them comes up heads?

Solution. We saw above that the probability of the complementary event - that is, that no coin comes up heads - is $\frac{1}{64}$. So the probability of at least one coin landing on heads is $1 - \frac{1}{64} = \boxed{\frac{63}{64}}$.

3 Symmetry in Probability

Taking advantage of symmetry in problems can save us a lot of unnecessary computation:

Example 7. A bag contains 100 tiles numbered with the positive integers 1 through 100. Van and Vex simultaneously draw a tile from the bag. What is the probability Van's tile number is larger than Vex's tile number?

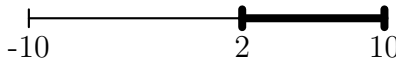
Solution. We *could* do extensive casework based on all the possible tiles Van (or Vex) could draw, but symmetry makes this problem simple: Let's say Van and Vex drew the tiles i and j in some order, where $i < j$. There's two ways for this to happen: Either Van draws tile i and Vex draws tile j , or Vex draws tile i and Van draws tile j . So for any two distinct tiles Van and Vex could draw, Van is just as likely to draw the higher-numbered tile as Vex is, and so the desired probability is $\boxed{\frac{1}{2}}$.

4 Geometric Probability

Some probability problems don't have quantities we can easily count. In this case, we can often turn to geometry as a useful tool: We can represent the possible outcomes as a geometric region - usually a length, area, or volume. Then, we can represent the successful outcomes as a portion of this region, and the probability this event occurs is the fraction of the size of the region we started with.

Example 8. A real number x is selected at random from the interval $[-10, 10]$. What is the probability that $|x| \geq |x - 4|$?

Solution. We can work with the geometric region given in the problem - the "possible region" is the interval $[-10, 10]$, and the "successful region" is the length of the interval inside our possible region with all x where $|x| \geq |x - 4|$. This condition means that the distance from x to 0 is greater than or equal to the distance from x to 4. This is true for all $x \geq 2$, so the successful region is $[2, 10]$.

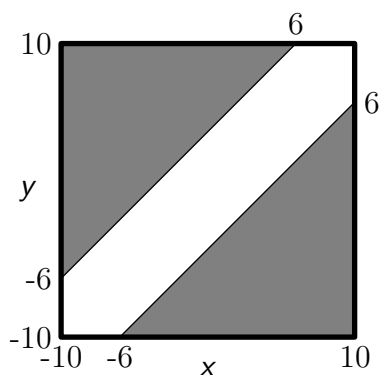


Our successful region has length 8 and the possible region has length 20, so the desired probability is $\frac{8}{20} = \boxed{\frac{2}{5}}$.

The geometric regions we can work with are not always as clear as they were in the last example, though.

Example 9. Two real numbers x, y are selected at random from the interval $[-10, 10]$. What is the probability that $|x - y| \geq 4$?

Solution. We can work with the geometric region given by the square with vertices $(-10, -10), (-10, 10), (10, 10), (10, -10)$, where a point (x, y) inside the square corresponds to a choice of $-10 \leq x \leq 10$ and $-10 \leq y \leq 10$. The shaded portion of the diagram below indicates all points where $|x - y| \geq 4$:



The two shaded triangles have coordinates $(-6, -10), (10, 6), (10, -10)$, and $(-10, -6), (6, 10), (-10, 10)$ respectively. Each of these triangles has area $\frac{1}{2} \cdot 16 \cdot 16 = 128$, so the total shaded area is 256. Also, the possible region has area $20 \cdot 20 = 400$. Therefore, the desired probability is $\frac{256}{400} = \boxed{\frac{16}{25}}$.



1. A pair of standard 6-sided fair dice is rolled once. The sum of the numbers rolled determines the diameter of a circle. What is the probability that the numerical value of the area of the circle is less than the numerical value of the circle's circumference?
2. Two (not necessarily distinct) integers are selected independently at random from the set $\{-9, -8, \dots, 9, 10\}$. What is the probability that the product of those numbers is greater than zero?
3. Four fair six-sided dice are rolled. What is the probability that at least three of the four dice show the same value?
4. A frog sitting at the point $(1, 2)$ begins a sequence of jumps, where each jump is parallel to one of the coordinate axes and has length 1, and the direction of each jump (up, down, right, or left) is chosen independently at random. The sequence ends when the frog reaches a side of the square with vertices $(0, 0)$, $(0, 4)$, $(4, 4)$, and $(4, 0)$. What is the probability that the sequence of jumps ends on a vertical side of the square?
5. A red ball and a green ball are randomly and independently tossed into bins numbered with positive integers so that for each ball, the probability that it is tossed into bin k is 2^{-k} for $k = 1, 2, 3, \dots$. What is the probability that the red ball is tossed into a higher-numbered bin than the green ball?
6. The numbers $1, 2, \dots, 9$ are randomly placed into the 9 squares of a 3×3 grid. Each square gets one number, and each of the numbers is used once. What is the probability that the sum of the numbers in each row and each column is odd?
7. Omar has four fair standard six-sided dice. Omar invented a game where he rolls his four dice over and over again until the number 1 does not appear on the top face of any of the dice. Omar wins the game if on that roll the top faces of his dice show at least one 2 and at least one 5. What is the probability Omar wins the game?



1. Phil parks his car on Riemann Road from 10:00 to 1:00, despite signs for two-hour parking only. If traffic patrol marks all the cars at a random time between 8:00 and 3:00 and checks in again two hours later to ticket all the cars still parked there, then what is the probability that Phil gets a ticket?
2. Chloe chooses a real number uniformly at random from the interval $[0, 2020]$. Independently, Laurent chooses a real number uniformly at random from the interval $[0, 4040]$. What is the probability that Laurent's number is greater than Chloe's number?
3. A point is chosen at random within the square in the coordinate plane whose vertices are $(0, 0)$, $(2020, 0)$, $(2020, 2020)$, and $(0, 2020)$. The probability that the point is within d units of a lattice point is $\frac{1}{2}$. (A point (x, y) is a lattice point if x and y are both integers.) What is d ?
4. Two points on the circumference of a circle of radius r are selected independently and at random. From each point a chord of length r is drawn in a clockwise direction. What is the probability that the two chords intersect?
5. Real numbers between 0 and 1, inclusive, are chosen in the following manner. A fair coin is flipped. If it lands heads, then it is flipped again and the chosen number is 0 if the second flip is heads and 1 if the second flip is tails. On the other hand, if the first coin flip is tails, then the number is chosen uniformly at random from the closed interval $[0, 1]$. Two random numbers x and y are chosen independently in this manner. What is the probability that $|x - y| > \frac{1}{2}$?
6. Real numbers x and y are chosen independently and uniformly at random from the interval $[0, 1]$. What is the probability that x , y , and 1 are the side lengths of an obtuse triangle?
7. Let S be a square of side length 1. Two points are chosen independently at random on the sides of S . What is the probability that the straight-line distance between the points is at least $\frac{1}{2}$?

1 Divisors

When dealing with the divisors of a number, algebraic factorizations can be useful.

Example. Find the greatest integer that divides every number of the form $n^3 - n$, where $n \geq 2$.

Solution. The first few values of $n^3 - n$ are

$$6, 24, 60, 120, 210, \dots$$

It looks like 6 divides every one of these numbers, so let's try proving that the answer is 6. (And since the first number is 6, this must be the largest such number.)

We can factor $n^3 - n$ as

$$n^3 - n = n(n^2 - 1) = n(n - 1)(n + 1).$$

Note that $n - 1$, n , and $n + 1$ are three consecutive integers. One of $n - 1$ and n must be divisible by 2, and one of $n - 1$, n , $n + 1$ must be divisible by 3. Therefore, their product $(n - 1)n(n + 1)$ must be divisible by $2 \cdot 3 = 6$. ■

Next, we will develop a formula for the number of divisors of an integer.

Example. Find the number of divisors of 360.

Solution. First, we write the prime factorization of 360, which is $360 = 2^3 \cdot 3^2 \cdot 5$. Every divisor of this number is of the form

$$2^a \cdot 3^b \cdot 5^c,$$

where $0 \leq a \leq 3$, $0 \leq b \leq 2$, and $0 \leq c \leq 1$. There are 4 possible values of a , 3 possible values of b , and 2 possible values of c , so 360 has $4 \cdot 3 \cdot 2 = 24$ divisors. ■

More generally, suppose a number has a prime factorization of

$$p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}.$$

Then every divisor of this number is of the form

$$p_1^{f_1} p_2^{f_2} \cdots p_k^{f_k},$$

where $0 \leq f_i \leq e_i$ for all i . For each i , there are $e_i + 1$ possible values of f_i , so the number has $(e_1 + 1)(e_2 + 1) \cdots (e_k + 1)$ divisors.

Exercise. A divisor of $12!$ is chosen at random. Find the probability that this divisor is a perfect square. (2020 AMC 10A, #15)

Exercise. Find the smallest positive integer that has exactly 16 divisors.

2 Decimal Representation

When dealing with the decimal representation of a number, it is often useful to write its decimal expansion. For example, we can write the three-digit number $\overline{XYZ}$ as $100X + 10Y + Z$.

Example. Find the smallest positive six-digit integer of the form $\overline{ABAABA}$, where A and B are distinct digits, such that the integer can be written as a product of exactly six primes (not necessarily distinct).

Solution. We can write

$$\begin{aligned}\overline{ABAABA} &= 100000A + 10000B + 1000A + 100A + 10B + A \\ &= 101101A + 10010B.\end{aligned}$$

This factors as

$$1001(101A + 10B) = 7 \cdot 11 \cdot 13 \cdot (101A + 10B).$$

We can also write $101A + 10B$ as $\overline{ABA}$. So, we are looking for the smallest three-digit number of the form $\overline{ABA}$ (where $A \neq B$), which is the product of exactly three primes (not necessarily distinct).

Checking, we find the smallest such number is $171 = 3^2 \cdot 19$, so the answer is 171171. ■

Example. A positive integer is called *lucky* if it is equal to 19 times the sum of its digits. Find all lucky positive integers.

Solution. Let n be the positive integer, let s be the sum of the digits of n , so $n = 19s$.

Let d be the number of digits in n . The smallest number that has d digits is 10^{d-1} , so $n \geq 10^{d-1}$. Also, each digit in n is at most 9, so $s \leq 9d$. Hence,

$$10^{d-1} \leq n = 19s \leq 19(9d) = 171d.$$

Thus, d must satisfy $10^{d-1} \leq 171d$. We list the first few values of 10^{d-1} and $171d$ in the table below:

d	10^{d-1}	$171d$
1	1	171
2	10	342
3	100	513
4	1000	684
5	10000	855

We see that $10^{d-1} \leq 171d$ for $d = 1, 2$, and 3 , but not for 4 and 5 . For larger values, 10^{d-1} grows faster than $171d$ (we omit the proof here), so the only possible values of d are $1, 2$, and 3 .

Since $n = 19s$, n must contain at least two digits. For $d = 2$, we can let $n = \overline{AB} = 10A + B$, and the condition $n = 19s$ becomes

$$10A + B = 19(A + B).$$

This simplifies to $9A + 18B = 0$, which has no solutions.

The only case left is where $d = 3$, which means n is a three-digit number. Let $n = \overline{ABC}$. Then the equation $n = 19s$ becomes

$$100A + 10B + C = 19(A + B + C).$$

This simplifies to $81A = 9B + 18C$. We can divide both sides by 9, to get

$$9A = B + 2C.$$

Since $B \leq 9$ and $C \leq 9$, $9A = B + 2C \leq 27$, so $A \leq 3$.

If $A = 1$, then $B + 2C = 9$. The possible pairs (B, C) are $(1, 4)$, $(3, 3)$, $(5, 2)$, $(7, 1)$, and $(9, 0)$, which gives us 114, 133, 152, 171, and 190 as possible values of n .

If $A = 2$, then $B + 2C = 18$. The possible pairs (B, C) are $(0, 9)$, $(2, 8)$, $(4, 7)$, $(6, 6)$, and $(8, 5)$, which gives us 209, 228, 247, 266, and 285 as possible values of n .

If $A = 3$, then $B + 2C = 27$. The only possible pair (B, C) is $(9, 9)$, which gives us $n = 399$.

So, there are a total of $5 + 5 + 1 = 11$ lucky numbers. ■

Exercise. The number 35 has the curious property that it is one more than the sum of the squares of its digits. In other words, $35 = 3^2 + 5^2 + 1^2$. Find the only other two-digit number with this property. (Mandelbrot, 2002)

Exercise. Find the four-digit number $\overline{ABCD}$ that satisfies $9 \times \overline{ABCD} = \overline{DCBA}$.



-

4. Find the smallest positive integer N with the following property: of the three numbers N , $N + 1$, and $N + 2$, one of them is divisible by 2^2 , one of them is divisible by 3^2 , one is divisible by 5^2 , and one is divisible by 7^2 .
5. Let S be a set of 6 integers taken from $\{1, 2, \dots, 12\}$ with the property that if a and b are elements of S with $a < b$, then b is not a multiple of a . What is the least possible value of an element in S ?
6. For some positive integer n , the number $110n^3$ has 110 positive integer divisors, including 1 and the number $110n^3$. How many positive integer divisors does the number $81n^4$ have?



1. The 5-digit number $\overline{A986B}$ is divisible by 72. What are A and B ?
2. In 1984, the engineer and prolific prime-finder Harvey Dubner found the largest known prime each of whose digits is either a one or a zero. The prime can be expressed as

$$\frac{10^{641} \times (10^{640} - 1)}{9} + 1.$$

How many digits does this prime have?

3. The digits 1, 2, 3, 4, 5, and 6 are each used once to compose a six-digit number $\overline{ABCDEF}$, such that the three-digit number $\overline{ABC}$ is divisible by 4, $\overline{BCD}$ is divisible by 5, $\overline{CDE}$ is divisible by 3, and $\overline{DEF}$ is divisible by 11. Find the six-digit number.
4. I have a three-digit number, and I add its digits to create its digit sum. When the digit sum of my number is subtracted from my number, the result is the square of the digit sum. What is my three-digit number?
5. Suppose

$$N = 1 + 11 + 101 + 1001 + 10001 + \cdots + \underbrace{1\,000\ldots0000\,1}_{50 \text{ zeros}}.$$

When N is calculated and written as a single integer, what is the sum of its digits?

6. A positive integer has the property that when its last digit is moved to the front of the number, the result is equal to four times the original number. Find the smallest positive integer that has this property.



1. What is the surface area of a cube whose volume is 27 cm^3 ?
(A) 6 cm^2 (B) 27 cm^2 (C) 36 cm^2
(D) 54 cm^2 (E) 144 cm^2

1.

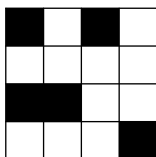
2. A fuse burns at a rate of exactly $2\frac{2}{3}$ centimeters per minute. How many minutes will it take to burn a 5 cm fuse?
(A) $\frac{8}{15}$ (B) $1\frac{7}{8}$ (C) 2 (D) $2\frac{1}{7}$ (E) $13\frac{1}{3}$

2.

3. Half of x is one less than twice x . What is the value of x ?
(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{4}{5}$

3.

4. What is the smallest number of squares that must be colored black to give the diagram below a single line of reflectional symmetry?



4.

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

5. Which of the answer choices could be used to make the statement below true?

The letter “e” appears exactly _____ times in this sentence.

- (A) ten (B) eleven (C) twelve
(D) thirteen (E) fourteen

5.



6. What is the smallest positive integer n for which $n/3$ is a perfect square, and $n/2$ is a perfect cube?
 (A) 216 (B) 324 (C) 432 (D) 864 (E) 1296

6.

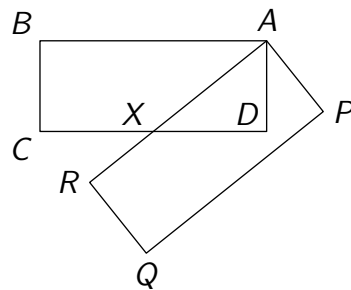
7. Trapezoid $ABCD$ has right angles A and B , with $AB = 3$, $BC = 4$, and $CD = 5$. What is the area of trapezoid $ABCD$?
 (A) 9 (B) 12.5 (C) 18 (D) 24 (E) 36

7.

8. What is the value of $\frac{1}{3} - \frac{1}{3^2} + \frac{1}{3^3} - \frac{1}{3^4} + \frac{1}{3^5} - \frac{1}{3^6} + \dots$?
 (A) 0 (B) $\frac{1}{6}$ (C) $\frac{1}{5}$ (D) $\frac{1}{4}$ (E) $\frac{1}{3}$

8.

9. In the diagram below, rectangles $ABCD$ and $APQR$ are congruent. Point X is the midpoint of $\overline{CD}$. If the area of triangle ADX is 20 cm^2 , what is the area of hexagon $APQRXD$?



- (A) 40 cm^2 (B) 50 cm^2 (C) 56 cm^2
 (D) 60 cm^2 (E) 80 cm^2

9.

10. At Vinny's Pizza, you can order a medium pizza with any combination of the 8 available toppings (including a pizza with no toppings at all). How many different medium pizzas can be ordered from Vinny's?
 (A) 8 (B) 9 (C) 36 (D) 64 (E) 256

10.

11. Regular hexagon $ABCDEF$ has sides of length 4. What is the area of rectangle $BCEF$?

(A) $8\sqrt{3}$ (B) 16 (C) $16\sqrt{3}$ (D) 36 (E) $24\sqrt{3}$

11.

12. A segment graphed on the coordinate plane has its endpoints at $(3, a)$ and $(21, -4b)$. The midpoint of the segment is at $(2a, b)$. What is the value of $a + b$?

(A) -1 (B) 1 (C) 3 (D) 5 (E) 7

12.

13. In a board game with 100 squares, on each turn you are allowed to move either 5 squares forward, 13 squares forward, or 1 square backwards. What is the fewest number of moves it will take for you to move from the 3rd to the 50th square?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

13.

14. Two numbers a and b satisfy $a + b = 20$ and $\frac{1}{a} + \frac{1}{b} = \frac{1}{2}$. What is the value of $a^2b + ab^2$?

(A) 80 (B) 200 (C) 400 (D) 640 (E) 800

14.

15. What is the sum of all solutions to the following equation?

$$x + 2 = 3\sqrt{x}.$$

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

15.

16. Each of the integers 2, 3, 13, 14, 22, and 23 labels one vertex of a regular hexagon such that the sum of the numbers on any two adjacent vertices is a perfect square. What number is on the vertex directly opposite the vertex labeled 23?

(A) 2 (B) 3 (C) 13 (D) 14 (E) 22

16.

17. Three people are selected at random. What is the probability that no two were born on the same day of the week? (Assume that a birthday can occur on any day of the week with equal probability.)

(A) $\frac{30}{49}$ (B) $\frac{3}{7}$ (C) $\frac{19}{49}$ (D) $\frac{5}{7}$ (E) $\frac{6}{7}$

17.

18. Sam always drives the same way to work. If Sam leaves for work at 7am and averages 40 miles per hour, he will arrive 10 minutes late. If he leaves at 7am and averages 50 miles per hour, he will arrive exactly on time. One morning, Sam left for work at 7am and arrived 5 minutes early. What was Sam's average speed in miles per hour that morning?

(A) 55 (B) $56\frac{1}{3}$ (C) $57\frac{1}{7}$ (D) $58\frac{1}{4}$ (E) 60

18.

19. Both a and n are positive integers such that $\frac{144}{a} = n$. What is the sum of all possible values of n ?

(A) 352 (B) 373 (C) 377 (D) 403 (E) 415

19.

20. How many of the distinct arrangements of the nine letters below include the sequence of letters in order that form the word "TEN"? The arrangement given is one such arrangement.

TWENTYTEN

(A) 2400 (B) 2460 (C) 4920 (D) 4980 (E) 5040

20.

1 Modular Arithmetic

For integers a and b , and a positive integer m , we say that a and b are *congruent modulo m* if $a - b$ is a multiple of m . We write this as $a \equiv b \pmod{m}$. For example, $138 \equiv 3 \pmod{5}$ because $138 - 3 = 135$ is a multiple of 5.

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

- $a + c \equiv b + d \pmod{m}$,
- $a - c \equiv b - d \pmod{m}$, and
- $ac \equiv bd \pmod{m}$.

It follows that if $a \equiv b \pmod{m}$, then

$$a^n \equiv b^n \pmod{m}$$

for any positive integer n .

Example. Find the number of positive integers n , $1 \leq n \leq 100$, such that $n^2 + 3n + 1$ is divisible by 5.

Solution. If $n \equiv 0 \pmod{5}$, then $n^2 + 3n + 1 \equiv 0^2 + 3 \cdot 0 + 1 \equiv 1 \pmod{5}$. We can handle all the cases $n \equiv r \pmod{5}$, where $0 \leq r \leq 4$, similarly.

$n \pmod{5}$	$n^2 + 3n + 1 \pmod{5}$
0	1
1	0
2	1
3	4
4	4

We see that $n^2 + 3n + 1 \equiv 0 \pmod{5}$ only when $n \equiv 1 \pmod{5}$. The numbers n in the range $1 \leq n \leq 100$ that work are then 1, 6, 11, $\dots$, 96, and there are 20 such numbers. ■

Looking for patterns using modular arithmetic can be an effective technique, especially when dealing with large numbers.

Example. First a is chosen at random from the set $\{1, 2, 3, \dots, 99, 100\}$, and then b is chosen at random from the same set. Find the probability that $3^a + 7^b$ has units digit 8. (1990 AHSME, #18)

Solution. Since we are interested in the units digit of $3^a + 7^b$, we can consider 3^a and 7^b modulo 10. And to see how 3^a and 7^b behave modulo 10, we compute the first few values:

n	$3^n \pmod{10}$	$7^n \pmod{10}$
1	3	7
2	9	9
3	7	3
4	1	1
5	3	7
6	9	9
7	7	3
8	1	1

We see that both $3^n \pmod{10}$ and $7^n \pmod{10}$ are periodic, with period 4. We can be sure of this since $3^4 \equiv 7^4 \equiv 1 \pmod{10}$. For example,

$$3^{13} \equiv (3^4)^3 \cdot 3 \equiv 1^3 \cdot 3 \equiv 3 \pmod{10}.$$

Thus, there are $4 \cdot 4 = 16$ possible outcomes for the units digits of 3^a and 7^b . (And since 100 is divisible by 4, $3^a \pmod{10}$ has an equal chance of being 1, 3, 7, or 9, and similarly for $7^b \pmod{10}$).

We want the units digit of $3^a + 7^b$ to be 8, i.e. $3^a + 7^b \equiv 8 \pmod{10}$. If $3^a \equiv r \pmod{10}$ and $7^b \equiv s \pmod{10}$, then $3^a + 7^b \equiv 8 \pmod{10}$ when (r, s) is one of the three pairs (1, 7), (7, 1), or (9, 9). Hence, the probability that $3^a + 7^b \equiv 8 \pmod{10}$ is $\boxed{\frac{3}{16}}$. ■

Exercise. How many of the first 2018 numbers in the sequence 101, 1001, 10001, 100001, ... are divisible by 101? (2018 AMC 10B, #13)

Exercise. Let $k = 2008^2 + 2^{2008}$. What is the units digit of $k^2 + 2^k$? (2008 AMC 10A, #24)

2 Algebra in Number Theory

Using algebra effectively is an important technique in many number theory problems. The following factorizations are particularly useful:

- (Difference of Squares) $x^2 - y^2 = (x - y)(x + y)$
- (Difference of Cubes) $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$
- (Difference of n th powers) $x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + y^{n-1})$
- (Sum of Cubes) $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
- (Sum of n th powers, n odd) $x^n + y^n = (x + y)(x^{n-1} - x^{n-2}y + x^{n-3}y^2 - \cdots + y^{n-1})$

Example. Find all solutions to $a^2 - b^2 = 75$ in positive integers.

Solution. By difference of squares, we can write the equation as

$$(a + b)(a - b) = 75.$$

Thus, the product of $a + b$ and $a - b$ is 75, and we can easily list the possible values of $a + b$ and $a - b$. (Note that $a + b > a - b$.) For each possible value of $a + b$ and $a - b$, we can solve for a and b . For example, for $a + b = 75$ and $a - b = 1$, we can add the equations to get $2a = 76$, so $a = 38$. Then $b = a - 1 = 37$.

$a + b$	$a - b$	a	b
75	1	38	37
25	3	14	11
15	5	10	5

Thus, the solutions (a, b) are $\boxed{(38, 37), (14, 11), (10, 5)}$. ■

The expression $mn + m + n$ does not factor, but if we add 1, the resulting expression does factor as

$$mn + m + n + 1 = (m + 1)(n + 1).$$

Adding a constant to an expression like $mn + m + n$ in order to factor it is an instance of *Simon's Favorite Factoring Trick*.

Example. A rectangle has integer side lengths. If 3 units are added to the height and 2 units are added to the width, then the area of the rectangle is tripled. Find all possible dimensions of the original rectangle.

Solution. Let a be the height, and let b be the width. Then

$$(a + 3)(b + 2) = 3ab,$$

which simplifies to $2ab - 2a - 3b - 6 = 0$. We can factor the first two terms as

$$2a(b - 1) - 3b - 6 = 0.$$

Seeking another factor of $b - 1$, we can then write this equation as $2a(b - 1) - 3(b - 1) - 3 - 6 = 0$, so

$$(2a - 3)(b - 1) = 9.$$

Thus, the product of $2a - 3$ and $b - 1$ is 9. We can form a table.

$2a - 3$	$b - 1$	a	b
1	9	2	10
3	3	3	4
9	1	6	2

Therefore, the possible dimensions of the original rectangle are $2 \times 10, 3 \times 4, 6 \times 2$.

Exercise. The number 35 has the property that when both its digits are increased by 2, and then multiplied, the result is $5 \times 7 = 35$, equal to the original number. Find all two-digit numbers that have this property.

Exercise. If $abc + ab + bc + ca + a + b + c = 104$, and a, b , and c are positive integers, then find $a^2 + b^2 + c^2$.



1. What is the units digit of 13^{2003} ?
2. A sequence is defined by $a_1 = 1$ and $a_n = a_{n-1}^2 + 1$ for all $n \geq 2$. Find the units digit of a_{1000} .
3. What is the tens digit of $2015^{2016} - 2017$?
4. The positive integers are written in a table as shown:

1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32	33	34	35	36
				$\vdots$			$\vdots$				

What is the probability that a randomly chosen column of this table contains a perfect square?

5. Find the remainder when $111 \dots 1$ (with 1000 1's) is divided by 37.
6. A number m is randomly selected from the set $\{11, 13, 15, 17, 19\}$, and a number n is randomly selected from $\{1999, 2000, 2001, \dots, 2018\}$. What is the probability that m^n has a units digit of 1?
7. Let $N = 123456789101112 \dots 4344$ be the 79-digit number that is formed by writing the integers from 1 to 44 in order, one after the other. What is the remainder when N is divided by 45?

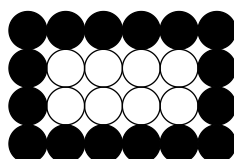


1. If p , q , and r are positive integers and

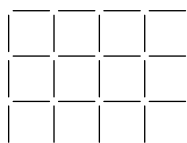
$$p + \frac{1}{q + \frac{1}{r}} = \frac{25}{19},$$

then what number does q equal?

2. In 1991 the population of a town was a perfect square. Ten years later, after an increase of 150 people, the population was 9 more than a perfect square. In 2011, with an increase of another 150 people, the population was once again a perfect square. What was the population of the town in 2011?
3. A border of black counters is placed around a rectangular array of white counters in a way similar to that shown in the diagram. If the number of white counters is equal to the number of black counters, for how many different numbers of white counters can this be done?



4. Find the largest positive integer n such that $n - 4$ divides n^2 .
5. As shown in the diagram, you can create a grid of squares 3 units high and 4 units wide using 31 matches. I would like to make a grid of squares a units high and b units wide, where $a < b$ are positive integers. Determine the sum of the areas of all such rectangles that can be made, each using exactly 337 matches.



6. Adding 1 to the product of four consecutive positive integers always results in a perfect square. The first 2017 such square numbers can be found:

$$1 \times 2 \times 3 \times 4 + 1 = 25 = 5^2$$

$$2 \times 3 \times 4 \times 5 + 1 = 121 = 11^2$$

$$3 \times 4 \times 5 \times 6 + 1 = 361 = 19^2$$

...

$$2017 \times 2018 \times 2019 \times 2020 + 1 = 16600254584281 = 4074341^2$$

In the list of 2017 numbers

$$5, 11, 19, \dots, 4074341,$$

whose squares are found in this way, how many have last digit equal to 1?

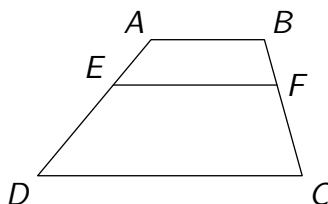




AMC10/12 Camp

Class 7 Handout 3: Math Battle Problems

1. In the diagram, $\overline{AB}$, $\overline{CD}$, and $\overline{EF}$ are parallel, with $AB = 3.6$, $CD = 13.2$, $AE = 2.2$, and $DE = 6.6$. Find the length EF .



2. Guy has a collection of 10 rulers, whose lengths are $1, 2, 4, 8, \dots, 512$, in meters. He arranges the 10 rulers in a line, in random order. What is the probability that there is a consecutive block of rulers whose lengths add up to 1000 meters?
3. Find all values of k for which the system of equations

$$\begin{aligned} kx + y + z &= k, \\ x + ky + z &= k, \\ x + y + kz &= k \end{aligned}$$

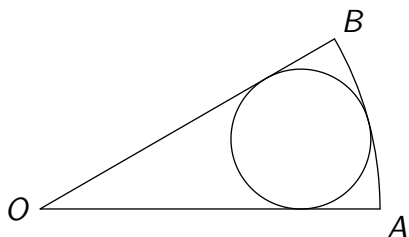
does *not* have a unique solution in x , y , and z .

4. A mathematical test consists of 90 questions. For each question, you get 4 points for a correct answer, -2 points for an incorrect answer, and 0 points for leaving the question unanswered. Find the number of different possible values of the total score.
5. Let S be a subset of $\{1, 2, 3, \dots, 2000\}$, so that for any two different elements a and b in S , $a + b$ is *not* divisible by 7. What is the largest possible number of elements in S ?
6. Find all solutions to

$$a^2 + 5a + 19 = b^2$$

where a and b are positive integers.

7. Points A and B lie on a circle centered at O with radius 9. A circle is inscribed in circular sector OAB with radius 1. Find the length of chord $\overline{AB}$.

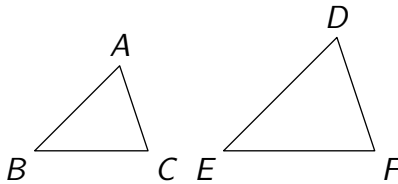


8. The 42 points $P_1, P_2, \dots, P_{42}$ lie on a straight line, in that order, so that the distance between P_n and P_{n+1} is $\frac{1}{n}$ for all $1 \leq n \leq 41$. What is the sum of the distances between every pair of these points? (Each pair of points is counted only once.)

1 Similar Triangles

Triangles ABC and DEF are *similar* if their corresponding angles are equal, i.e. $\angle A = \angle D$, $\angle B = \angle E$, and $\angle C = \angle F$. To show that two triangles are similar, we only need to establish two equal pairs, like $\angle A = \angle D$ and $\angle C = \angle F$, since the angles of a triangle add up to 180° .

Note: When we write similar triangles, the vertices must be in corresponding order. For example, if triangles ABC and DEF are similar, then we can also write that triangles CAB and FDE are similar, but we cannot write that triangles CAB and DEF are similar.

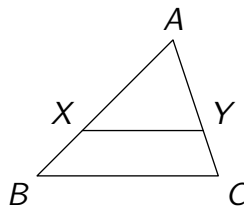


In similar triangles, the ratio of corresponding sides are the same, i.e.

$$\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}.$$

Furthermore, all corresponding parts have the same ratio. For example, if h is the altitude from A to $\overline{BC}$, and h' is the altitude from D to $\overline{EF}$, then $\frac{h}{h'} = \frac{BC}{EF}$.

Suppose X and Y are on sides $\overline{AB}$ and $\overline{AC}$ of triangle ABC , such that $\overline{XY}$ is parallel to $\overline{BC}$.

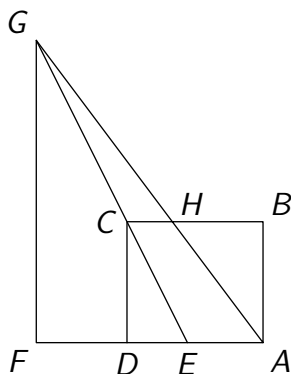


Then $\angle AXY = \angle ABC$ and $\angle AYX = \angle ACB$, so triangles AXY and ABC are similar. Furthermore, the ratio of their areas is equal to the square of the ratio of corresponding sides. For example,

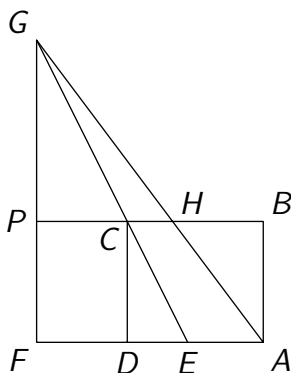
$$\frac{[AXY]}{[ABC]} = \left(\frac{XY}{BC}\right)^2,$$

where $[T]$ stands for the area of triangle T . Similar triangles often arise from parallel lines.

Example. In rectangle $ABCD$, we have $AB = 8$, $BC = 9$, H is on $\overline{BC}$ with $BH = 6$, E is on $\overline{AD}$ with $DE = 4$, line EC intersects line AH at G , and F is on line AD with $\overline{GF} \perp \overline{AF}$. Find the length GF . (2003 AMC 10 A, #22)



Solution. Let lines BC and FG intersect at P .



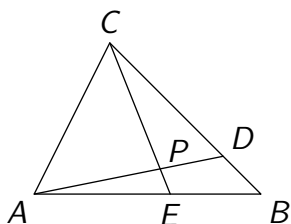
Let $x = GP$. Since $\overline{CH}$ and $\overline{EA}$ are parallel, triangles GCH and GEA are similar. In similar triangles, corresponding parts have the same ratio, so

$$\frac{GP}{GF} = \frac{CH}{EA},$$

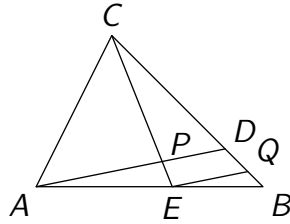
which gives us $\frac{x}{x+8} = \frac{3}{5}$. Solving for x , we find $x = 12$. Therefore, $GF = x + 8 = \boxed{20}$. ■

If no similar triangles are present in a diagram, it can be useful to construct similar triangles by drawing parallel lines, especially if the problem involves ratios of lengths.

Example. In triangle ABC , lines CE and AD are drawn so that $CD/DB = 3$ and $AE/EB = 3/2$. Find CP/PE , where P is the intersection point of $\overline{CE}$ and $\overline{AD}$. (1963 AHSME, #39)



Solution. Let the line through E parallel to $\overline{AD}$ intersect $\overline{BC}$ at Q .



Then triangles BEQ and BAD are similar, so

$$\frac{BQ}{QD} = \frac{BE}{EA} = \frac{2}{3}.$$

Also, triangles CPD and CEQ are similar, so

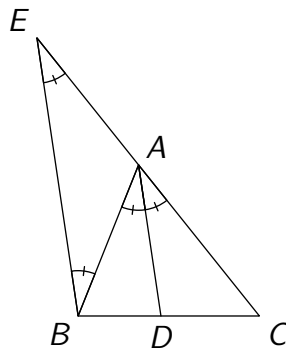
$$\frac{CP}{PE} = \frac{CD}{DQ}.$$

Note that $CD = 3 \cdot DB$ and $DQ = \frac{3}{5} \cdot DB$, so $\frac{CP}{PE} = \frac{CD}{DQ} = \frac{3 \cdot DB}{\frac{3}{5} \cdot DB} = \boxed{5}$. ■

Example. (Angle Bisector Theorem) In triangle ABC , $\overline{AD}$ bisects $\angle BAC$. Show that

$$\frac{BD}{CD} = \frac{AB}{AC}.$$

Solution. We apply the same strategy of drawing parallel lines, in order to construct similar triangles. Here, we draw the line through B parallel to $\overline{AD}$, and let it intersect line AC at E .



Note that $\angle BEA = \angle DAC = \angle DAB = \angle EBA$, so triangle ABE is isosceles with $AE = AB$. Also, triangles EBC and ADC are similar, so

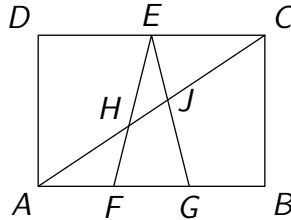
$$\frac{BC}{CD} = \frac{EC}{AC}.$$

We can write this as

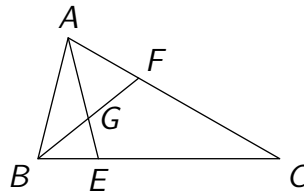
$$\frac{BD + CD}{CD} = \frac{EA + AC}{AC} = \frac{AB + AC}{AC}.$$

Hence, $\frac{BD}{CD} = \frac{AB}{AC}$. ■

Exercise. In rectangle $ABCD$, points F and G lie on AB so that $AF = FG = GB$ and E is the midpoint of $\overline{DC}$. Also, $\overline{AC}$ intersects $\overline{EF}$ at H and $\overline{EG}$ at J . The area of rectangle $ABCD$ is 70. Find the area of triangle EHJ . (2001 AMC 12, #22)



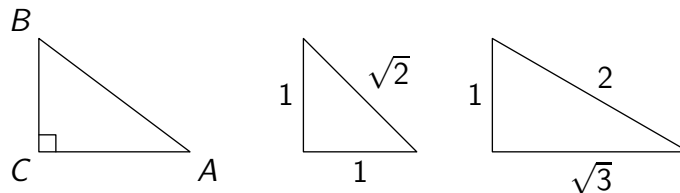
Exercise. In triangle ABC , F divides side $\overline{AC}$ in the ratio 1 : 2. Let E be the point of intersection of side $\overline{BC}$ and $\overline{AG}$, where G is the midpoint of $\overline{BF}$. Find $BE : EC$. (1971 AHSME, #26)



Exercise. Let $ABCD$ be a parallelogram. Extend $\overline{DA}$ through A to a point P , and let $\overline{PC}$ meet $\overline{AB}$ at Q and $\overline{DB}$ at R . Given that $PQ = 735$ and $QR = 112$, find RC . (1998 AIME, #6)

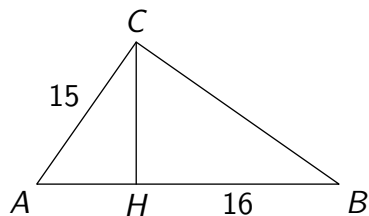
2 The Pythagorean Theorem

If triangle ABC is a right triangle with $\angle ACB = 90^\circ$, then the Pythagorean Theorem asserts that $AC^2 + BC^2 = AB^2$. Special right triangles include the 45° - 45° - 90° (where the sides are in the ratio $1 : 1 : \sqrt{2}$), and the 30° - 60° - 90° triangle (where the sides are in the ratio $1 : \sqrt{3} : 2$).

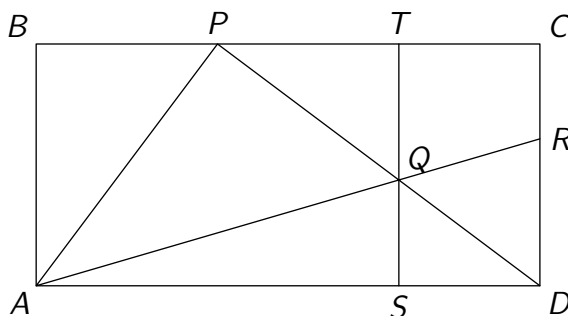




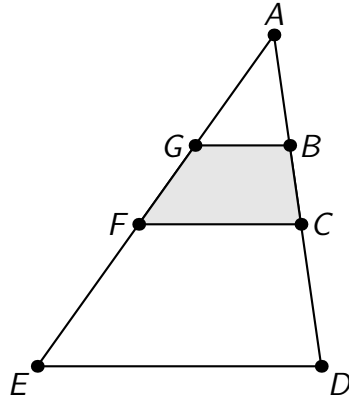
1. In triangle ABC , $AB = 8$, $BC = 7$, $CA = 6$, and side $\overline{BC}$ is extended past C to a point P so that triangle PAB is similar to triangle PCA . Find the length of PC .
2. A right triangle ABC with hypotenuse $\overline{AB}$ has side $AC = 15$. Altitude $\overline{CH}$ divides $\overline{AB}$ into segments $\overline{AH}$ and $\overline{HB}$, with $HB = 16$. Find the area of triangle ABC .



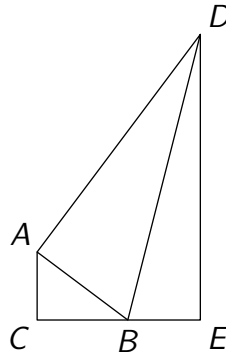
3. In triangle ABC with a right angle at C , point D lies in the interior of $\overline{AB}$ and point E lies in the interior of $\overline{BC}$ so that $AC = CD$, $DE = EB$, and the ratio $AC : DE = 4 : 3$. What is the ratio $AD : DB$?
4. In rectangle $ABCD$ with $AB = 16$, P is a point on $\overline{BC}$ so that $\angle APD = 90^\circ$. $\overline{TS}$ is perpendicular to $\overline{BC}$ with $BP = PT$, as shown. $\overline{PD}$ intersects $\overline{TS}$ at Q . Point R is on $\overline{CD}$ such that $\overline{RA}$ passes through Q . In triangle PQA , $PA = 20$, $AQ = 25$, and $QP = 15$. Find RD .



5. In triangle ADE , points B and C are on side $\overline{AD}$, and points F and G are on side $\overline{AE}$, so that $\overline{BG} \parallel \overline{CF} \parallel \overline{DE}$. The area of triangle ABG is 36, the area of trapezoid $CFED$ is 144, and $AB = CD$. Find the area of trapezoid $BGFC$.

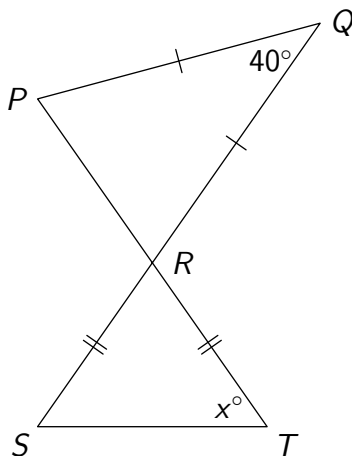


6. Triangle ABC has a right angle at C , $AC = 3$ and $BC = 4$. Triangle ABD has a right angle at A and $AD = 12$. Points C and D are on opposite sides of AB . The line through D parallel to $\overline{AC}$ meets line CB extended at E . Find DE/DB .

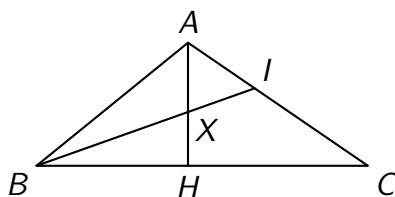




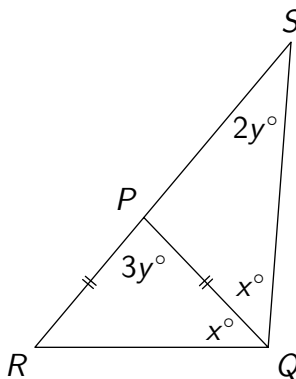
1. In the diagram, PRT and QRS are straight lines. What is the value of x ?



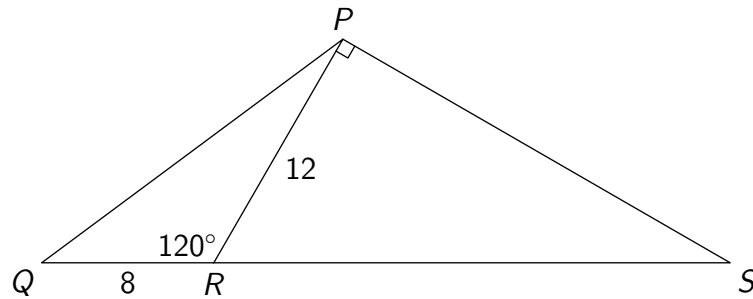
2. In triangle ABC , altitude $\overline{AH}$ intersects angle bisector $\overline{BI}$ at point X . If $\angle BAC = 117^\circ$ and $\angle ACB = 35^\circ$, then determine $\angle HXI$.



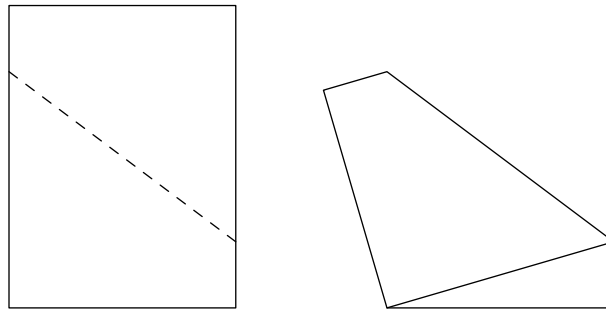
3. In the diagram, P is on $\overline{RS}$ so that $\overline{QP}$ bisects $\angle SQR$. Also, $PQ = PR$, $\angle RSQ = 2y^\circ$, and $\angle RPQ = 3y^\circ$. What is the measure, in degrees, of $\angle RPQ$?



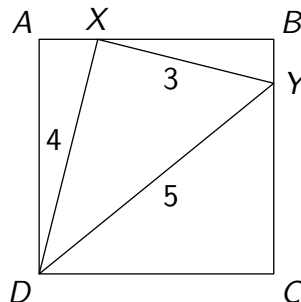
4. In the diagram, R is on $\overline{QS}$ and $QR = 8$. Also, $PR = 12$, $\angle PRQ = 120^\circ$, and $\angle RPS = 90^\circ$. What is the area of triangle QPS ?



5. A 6×8 piece of paper is folded over so that one corner goes to the opposite corner. Find the length of the crease.



6. In rectangle $ABCD$, E is a point on side $\overline{CD}$ so that triangle DBE is isosceles and triangle ABE is right. Compute AB/AD . Express your answer in the form $\sqrt{m + \sqrt{n}}$, where m and n are positive integers.
7. A triangle with side lengths 3, 4, and 5 is inscribed in a square, as shown below. Find the area of the square.





AMC10/12 Camp
Class 8 Handout 3: Mock AMC Contest

1. The product $5 \times 7 \times 11$ works out to 385. What is the value of $0.5 \times 0.7 \times 0.11$?
(A) 0.00385 (B) 0.0385 (C) 0.385 (D) 3.85 (E) 38.5

1.

2. Ernie can mow half of his lawn in three quarters of an hour. How many minutes will it take for him to mow two-thirds of his lawn at this rate?
(A) 50 (B) 60 (C) 75 (D) 90 (E) 120

2.

3. The tens digit of the product $1 \times 2 \times 3 \times \cdots \times 98 \times 99$ is
(A) 0 (B) 1 (C) 2 (D) 4 (E) 9

3.

4. If $a = 2b - 5$, then b is equal to
(A) $\frac{a}{2}$ (B) $\frac{a}{2} + 5$ (C) $\frac{a - 5}{2}$ (D) $\frac{a + 5}{2}$ (E) $2a + 5$

4.

5. Two coplanar circles intersect such that the area of their intersection is half the area of the smaller circle, and two-ninths the area of the larger circle. What is the ratio of the diameter of the smaller circle to the diameter of the larger one?

5.

- (A) $\frac{2}{3}$ (B) $\frac{6}{11}$ (C) $\frac{4}{9}$ (D) $\frac{\sqrt{66}}{11}$ (E) $\frac{\sqrt{14}}{7}$



6. The first term of an arithmetic sequence is 7. One of the terms in the sequence is 84, and the last term in the sequence is 139. If the terms of the sequence are non-consecutive integers, how many terms are in the sequence?

(A) 7 (B) 11 (C) 12 (D) 13 (E) 15

6.

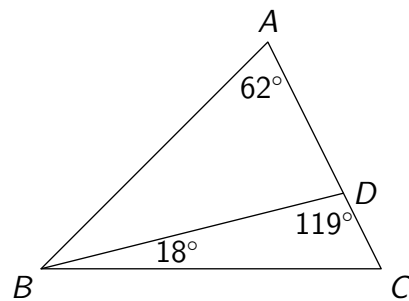
7. Compute $206^2 - 194^2$.

(A) 3200 (B) 3600 (C) 4000 (D) 4400 (E) 4800

7.

8. In triangle ABC , D lies on side $\overline{AC}$. Given $\angle A = 62^\circ$, $\angle CBD = 18^\circ$, and $\angle BDC = 119^\circ$, find $\angle ABD$.

8.



(A) 43° (B) 57° (C) 80° (D) 110° (E) 137°

9. An equilateral triangle has the property that the number of centimeters in its perimeter and the number of square centimeters in its area are equal. How many centimeters are in the length of the triangle's altitude?

(A) 3 (B) $2\sqrt{3}$ (C) $3\sqrt{3}$ (D) 6 (E) 12

9.

10. Tonya and Nancy start simultaneously at each end of a marked path on a skating rink, and skate towards each other along the path until they meet. Then, they turn around and skate back along the path. Tonya and Nancy skate at constant speeds, and Tonya skates slightly faster than Nancy. Who returns to her starting point first?

(A) Tonya (B) Nancy (C) They arrive at the same time
(D) It depends on how much faster Tonya is
(E) It depends on how long the path is

10.

11. How many integers between 100 and 1000 have the property that the sum of the digits is equal to 10?

(A) 36 (B) 54 (C) 55 (D) 62 (E) 63

11.

12. A coin with diameter $\frac{1}{2}$ is tossed onto a floor which is infinitely tiled by squares of side length 1. What is the probability that the coin lands entirely within one of the squares?

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{3}{4}$ (D) $\frac{\pi}{4}$ (E) $\frac{4}{\pi^2}$

12.

13. What is the sum of the first 50 positive even integers?

(A) 650 (B) 1275 (C) 2500 (D) 2550 (E) 5050

13.

14. Suppose f is a function such that $f(x+1, y) = f(x, y) + y + 1$, $f(x, 0) = x$, and $f(x, y) = f(y, x)$ for all x and y . Then $f(12, 5)$ is equal to

(A) 17 (B) 60 (C) 67 (D) 77 (E) 83

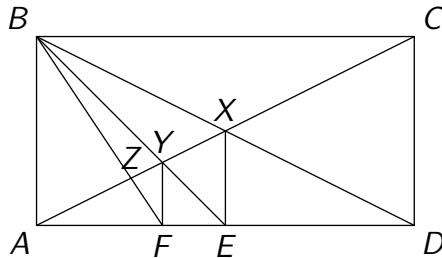
14.

15. Let $X = \{1, 2, 3, 4, 5, 6\}$. How many non-empty subsets of X do not contain two consecutive integers?

(A) 16 (B) 18 (C) 20 (D) 21 (E) 24

15.

16. Let $ABCD$ be a rectangle. Let $\overline{AC}$ and $\overline{BD}$ intersect at X , and let E be the foot of the perpendicular from X to $\overline{AD}$. Let $\overline{AC}$ and $\overline{BE}$ intersect at Y , and let F be the foot of the perpendicular from Y to $\overline{AD}$. Finally, let $\overline{AC}$ and $\overline{BF}$ intersect at Z . If $AB = 1$ and $AD = 2$, then ZF is equal to



- (A) $\frac{1}{3}$ (B) $\frac{\sqrt{2}}{5}$ (C) $\frac{3}{10}$ (D) $\frac{2\sqrt{5}}{15}$ (E) $\frac{\sqrt{13}}{12}$

17. Let k be the smallest positive integer such that the numbers $1 + k$, $1 + 2k$, and $1 + 8k$ are divisible by 3, 5, and 7, respectively. Then k lies in the interval
- (A) 0 to 20 (B) 21 to 40 (C) 41 to 60 (D) 61 to 80
(E) 81 to 100

18. Let n be the ten-digit number 9999999999. The sum of the digits of n^3 is
- (A) 99 (B) 108 (C) 180 (D) 199 (E) 297

19. For an integer n , let $T_n = \frac{1}{2}n(n+1)$. Find the number of ordered pairs of integers (m, n) such that $T_m - T_n = 10$.
- (A) 4 (B) 6 (C) 8 (D) 12 (E) 16

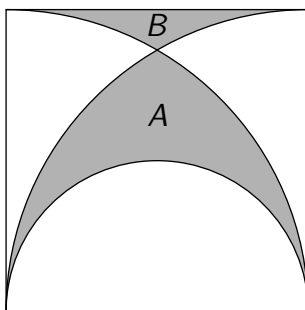
20. Fifteen lines are drawn in a plane, with no three concurrent and no two parallel, dividing the plane into 121 non-overlapping regions. Some of these regions are completely bounded by line segments and some are not. The number that are completely bounded by line segments is
- (A) 30 (B) 45 (C) 75 (D) 81 (E) 91

21. The smallest positive integer n such that $\sqrt{n} - \sqrt{n-1} < 0.01$ is
 (A) 2233 (B) 2335 (C) 2336 (D) 2500 (E) 2501

21.

22. The diagram shows a semicircle and two quarter circles inscribed in a square of side length 2. The difference between the area of the shaded region A and the area of the shaded region B equals

22.



- (A) $\frac{3}{2}\pi - 4$ (B) $\frac{1}{3}\pi - \frac{1}{3}$ (C) $\frac{3}{2} - \frac{1}{4}\pi$
 (D) $4 - \pi$ (E) $\frac{1}{4}\pi$

23. An acute triangle has side lengths a , b , and c , where $a < b < c$. A side of the triangle is chosen, and the triangle is rotated around the side in space, forming a solid of revolution. Which side should be chosen to form the solid with the greatest volume?
 (A) The side with length a (B) The side with length b
 (C) The side with length c (D) All solids have the same volume
 (E) There is not enough information in the problem

23.

24. A *perfect power* is a number of the form m^n , where m and n are positive integers greater than 1. How many positive integers less than or equal to 2^{12} can be expressed as a perfect power?
 (A) 72 (B) 81 (C) 82 (D) 85 (E) 87

24.

25. Let

$$x = \frac{2 \cdot 4 \cdot 6 \cdots 1998 \cdot 2000}{1 \cdot 3 \cdot 5 \cdots 1997 \cdot 1999}.$$

Then

- (A) $x < 10$ (B) $10 < x < 30$ (C) $30 < x < 50$
 (D) $50 < x < 70$ (E) $x > 70$

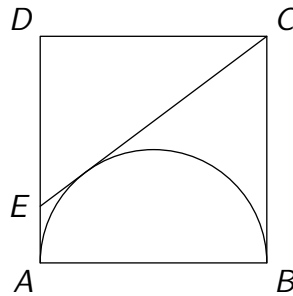
25.



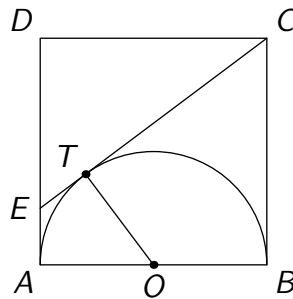
1 Circle Basics

When a circle is tangent to a line or another circle, it is generally useful to connect the center of the circle to a point of tangency by drawing a radius. Since this radius is perpendicular to the line of tangency, this will create a right angle that we can work with.

Example. Square $ABCD$ has side length 2. A semicircle with diameter $\overline{AB}$ is constructed inside the square and is tangent to the semicircle from C intersects side $\overline{AD}$ at E . What is the length of $\overline{CE}$? (2004 AMC 12 A, #18)



Solution. Let O be the midpoint of $\overline{AB}$, and let T be the point where the semicircle is tangent to $\overline{CE}$.



Since $\overline{EA}$ and $\overline{ET}$ are tangents from the same point to the same circle, $EA = ET$. Let $t = EA = ET$. Likewise, $CT = CB = 2$.

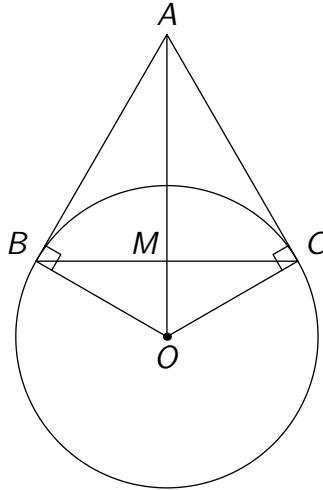
Then $EC = ET + TC = t + 2$, and $DE = AD - EA = 2 - t$. By the Pythagorean Theorem on right triangle EDC ,

$$(2 - t)^2 + 2^2 = (2 + t)^2.$$

Solving, we find $t = \frac{1}{2}$. Therefore, $CE = \boxed{\frac{5}{2}}$. ■

Example. Sides $\overline{AB}$ and $\overline{AC}$ of equilateral triangle ABC are tangent to a circle at points B and C , respectively. If the side length of triangle ABC is 2, then what is the area of the portion of triangle ABC that lies outside the circle?

Solution. Let O be the center of the circle. Then $\angle OBA = 90^\circ$ and $\angle BAO = \frac{\angle BAC}{2} = 30^\circ$, so triangle AOB is a 30° - 60° - 90° triangle. It follows that $\angle BOC = 120^\circ$, and $OB = \frac{2}{\sqrt{3}}$.



Let M be the midpoint of $\overline{BC}$. Then triangle BOM is a 30° - 60° - 90° triangle, so $MO = \frac{2/\sqrt{3}}{2} = \frac{1}{\sqrt{3}}$ and $BM = 1$, so the area of triangle BOM is $\frac{1}{2} \cdot 1 \cdot \frac{1}{\sqrt{3}} = \frac{1}{2\sqrt{3}}$. This means the area of triangle BOC is $\frac{1}{\sqrt{3}}$.

The area of circular sector BOC is $\frac{1}{3} \cdot \left(\frac{2}{\sqrt{3}}\right)^2 \cdot \pi = \frac{4}{9}\pi$, so the area of the circular segment cut off by $\overline{BC}$ is

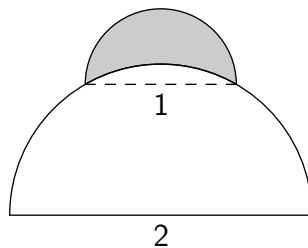
$$\frac{4}{9}\pi - \frac{1}{\sqrt{3}}.$$

Finally, the area of triangle ABC is $\frac{\sqrt{3}}{4} \cdot 2^2 = \sqrt{3}$, so the area of the portion of triangle ABC that lies outside the circle is

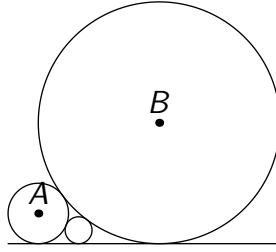
$$\sqrt{3} - \left(\frac{4}{9}\pi - \frac{1}{\sqrt{3}}\right) = \boxed{\frac{4}{3}\sqrt{3} - \frac{4}{9}\pi}.$$

■

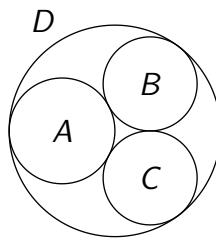
Exercise. A semicircle of diameter 1 sits at the top of a semicircle of diameter 2, as shown. The shaded area inside the smaller semicircle and outside the larger semicircle is called a *lune*. Determine the area of this lune. (2003 AMC 10 A, #19)



Exercise. A circle centered at A with a radius of 1 and a circle centered at B with a radius of 4 are externally tangent. A third circle is tangent to the first two and to one of their common external tangents as shown. Find the radius of the third circle. (2001 AMC 12, #18)

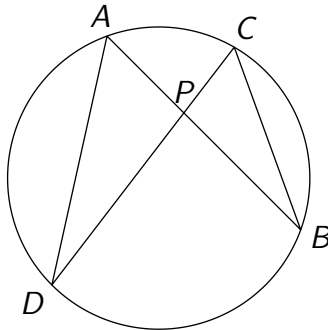


Exercise. Circles A , B and C are externally tangent to each other and internally tangent to circle D . Circles B and C are congruent. Circle A has radius 1 and passes through the center of D . What is the radius of circle B ? (2004 AMC 12A, #19)



2 Power of a Point

Let $\overline{AB}$ and $\overline{CD}$ be chords of a circle that intersect at P .

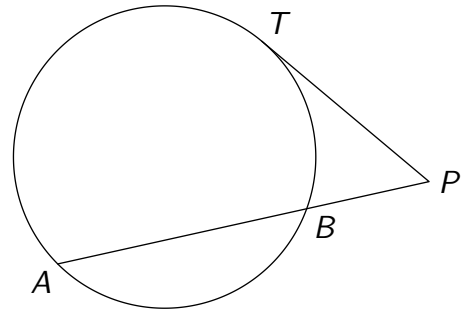
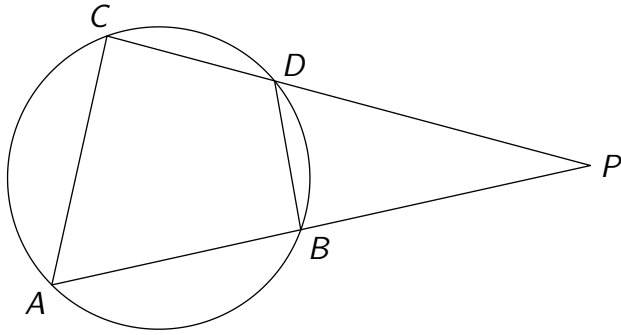


By the Inscribed Angle Theorem, $\angle DAB = \angle DCB$. Also, $\angle APD = \angle CPB$, so triangles APD and CPB are similar. Hence,

$$\frac{AP}{CP} = \frac{DP}{BP} \Rightarrow AP \cdot BP = CP \cdot DP.$$

This result tells us that for a fixed point P inside the circle, for any chord $\overline{AB}$ that passes through P , the quantity $AP \cdot BP$ is always the same. This quantity is called the *power of point P* with respect to the circle.

Similarly, if chords $\overline{AB}$ and $\overline{CD}$ when extended meet at a point P outside the circle, then the relationship $AP \cdot BP = CP \cdot DP$ still holds.

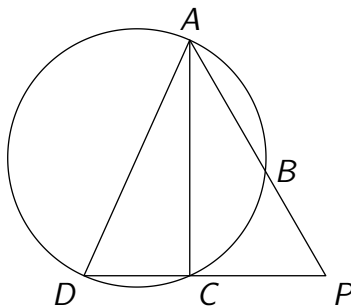


An interesting special case is when points C and D coincide at the same point T , so $\overline{PT}$ is tangent to the circle at T . In this case, the relationship becomes $AP \cdot BP = TP^2$.

These length relationships are most useful when we are given lengths relating to a circle, and we want to find other lengths.

Example. A circle of radius r has chords $\overline{AB}$ of length 10 and $\overline{CD}$ of length 7. When $\overline{AB}$ and $\overline{CD}$ are extended through B and C , respectively, they intersect at P , which is outside the circle. If $\angle APD = 60^\circ$ and $BP = 8$, then find r^2 . (1992 AHSME, #27)

Solution. Let $x = CP$. Then by power of a point, $AP \cdot BP = CP \cdot DP$, or $144 = x(x + 7)$. This simplifies to $x^2 + 7x - 144 = 0$, which factors as $(x - 9)(x + 16) = 0$. Hence, $x = 9$.



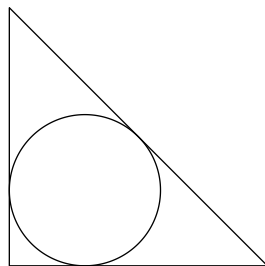
Since $\angle APC = 60^\circ$ and $AP = 2 \cdot CP$, triangle APC is a 30° - 60° - 90° triangle. Then $\angle ACD = 90^\circ$, which means that $\overline{AD}$ is a diameter of the circle. Then by the Pythagorean Theorem, $4r^2 = 7^2 + (9\sqrt{3})^2 = 292$, so $r^2 = \boxed{73}$. ■

Exercise In triangle ABC with altitude $\overline{AD}$, $\angle BAC = 45^\circ$, $DB = 3$, and $CD = 2$. Find the area of triangle ABC . (HMMT, 2005)

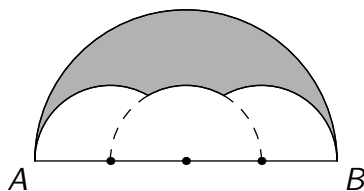
Exercise Triangle ABC and point P in the same plane are given. Point P is equidistant from A and B , angle APB is twice angle ACB , and $\overline{AC}$ intersects $\overline{BP}$ at point D . If $PB = 3$ and $PD = 2$, then find $AD \cdot CD$. (1997 AHSME, #26)



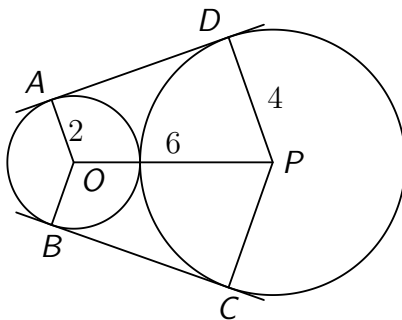
1. A unit circle is inscribed in an isosceles right triangle. Find the area of the triangle.



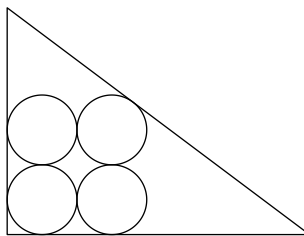
2. Three semicircles of radius 1 are constructed on diameter $\overline{AB}$ of a semicircle of radius 2. The centers of the small semicircles divide $\overline{AB}$ into four line segments of equal length, as shown. What is the area of the shaded region that lies within the large semicircle but outside the smaller semicircles?



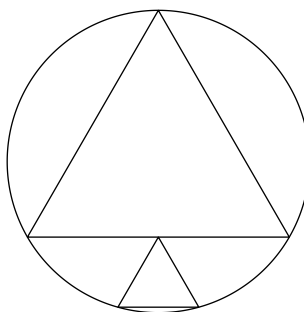
3. In triangle ABC , $BC = 15$, $AC = 13$, and $AB = 4$. A semicircle, centered on side $\overline{BC}$, is tangent to $\overline{AB}$ and $\overline{AC}$. Find the radius of the semicircle.
4. Circles with centers at O and P have radii 2 and 4, respectively, and are externally tangent. Points A and B on the circle with center O and points C and D on the circle with center P are such that $\overline{AD}$ and $\overline{BC}$ are common external tangents to the circles. What is the area of the concave hexagon $AOBCPD$?



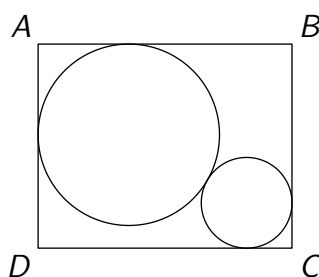
5. Four congruent circles are inscribed in a right triangle with sides 3, 4, and 5, as shown below. What is the common radius?



6. An equilateral triangle is inscribed in a circle. A second equilateral triangle is inscribed, as shown, so that its sides are parallel to the corresponding sides of the first equilateral triangle. Find the ratio of the side length of the large equilateral triangle to the side length of the small equilateral triangle.

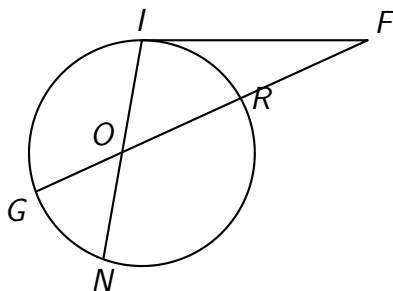


7. Two circles, with radii 3 and 5, are tangent to the sides of rectangle $ABCD$ and to each other, as shown below. The perimeter of the rectangle is 52. Find the area of the rectangle.

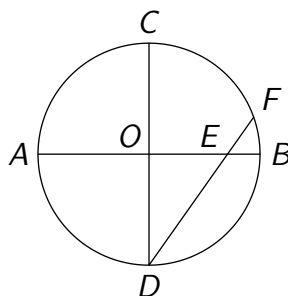




1. Let P be a point outside of circle O . A segment is drawn from P such that it is tangent to circle O at point T . Meanwhile, a secant from P intersects O at A and B , such that $PA < PB$. If $PA = 3$ and $PT = AB - PA$, then what is PB ?
2. Segment $\overline{IF}$ is tangent to the circle at point I . We are given that $IF = 21\sqrt{2}$, $IO = 20$, $ON = 12$, $RF = 18$, and $OR > GO$. Compute the length OR .



3. In the figure, $\overline{AB}$ and $\overline{CD}$ are diameters of the circle with center O , $\overline{AB} \perp \overline{CD}$, and chord $\overline{DF}$ intersects $\overline{AB}$ at E . If $DE = 6$ and $EF = 2$, then find the area of the circle.



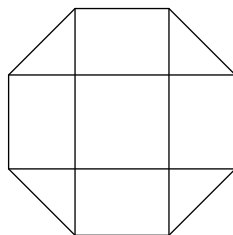
4. Let A, B, C , and D be points on a circle such that $AB = 11$ and $CD = 19$. Point P is on segment $\overline{AB}$ with $AP = 6$, and Q is on segment $\overline{CD}$ with $CQ = 7$. The line through P and Q intersects the circle at X and Y . If $PQ = 27$, find XY .
5. Let $ABCD$ be a quadrilateral inscribed in a circle, so that $\overline{BD}$ is a diameter of the circle. Let E be the projection of D onto $\overline{AC}$. If $AE = 8$, $DE = 7$, and $CE = 11$, then find the distance from B to chord $\overline{AC}$.
6. In triangle ABC , $AB = 15$, $AC = 6$, and $BC = 17$. The tangent to the circumcircle of triangle ABC at A meets $\overline{BC}$ at D . Find AD .



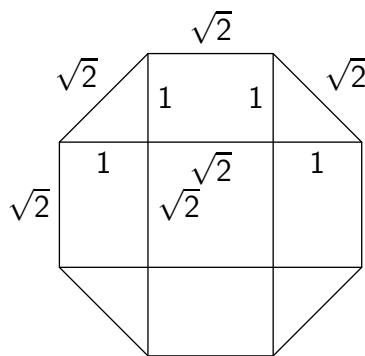
1 Polygons

One useful way to deal with polygons is to add lines in order to break down a polygon into simpler shapes, such as triangles or rectangles.

Example. A dart board is a regular octagon divided into regions as shown. Suppose that a dart thrown at the board is equally likely to land anywhere on the board. What is probability that the dart lands within the center square? (2011 AMC 10B, #16)



Solution. Each angle of a regular octagon is 135° , so each triangle in the diagram is a 45° - 45° - 90° triangle. We want to compute the probability that the dart lands in the center square, which is the area of the center square divided by the area of the whole octagon. Note that this probability does not change when we scale the diagram, so without loss of generality, we can let the sides of these triangles be 1, 1, and $\sqrt{2}$. Then the side length of the regular octagon is $\sqrt{2}$.



Adding up the elements of the regular octagon gives us a total area of

$$4 \cdot \frac{1}{2} \cdot 1 \cdot 1 + 4 \cdot 1 \cdot \sqrt{2} + \sqrt{2} \cdot \sqrt{2} = 4 + 4\sqrt{2}.$$

The area of the central square is $(\sqrt{2})^2 = 2$, so the probability we seek is

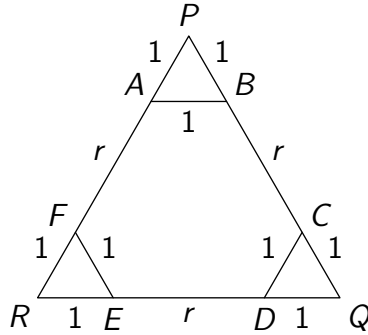
$$\frac{2}{4 + 4\sqrt{2}} = \frac{1}{2(\sqrt{2} + 1)} = \frac{\sqrt{2} - 1}{2(\sqrt{2} + 1)(\sqrt{2} - 1)} = \boxed{\frac{\sqrt{2} - 1}{2}}.$$

■



Example. Equiangular hexagon $ABCDEF$ has side lengths $AB = CD = EF = 1$ and $BC = DE = FA = r$. The area of triangle ACE is 70% of the area of the hexagon. What is the sum of all possible values of r ? (2010 AMC 10A, #19)

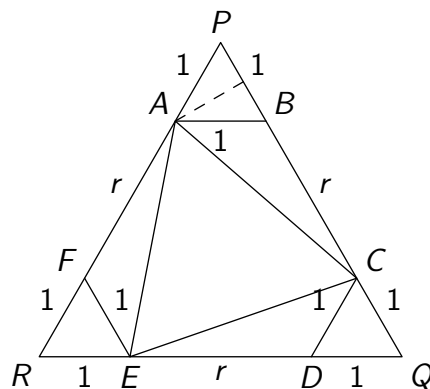
Solution. Since hexagon $ABCDEF$ is equiangular, every angle is equal to 120° . We can then extend the sides of the hexagon to form an equilateral triangle PQR .



We can compute the area of hexagon $ABCDEF$ by taking the area of equilateral triangle PQR , and subtracting the areas of triangles ABP , CDQ , and EFR . The area of an equilateral triangle with side length s is $\frac{\sqrt{3}}{4}s^2$, so

$$[ABCDEF] = \frac{\sqrt{3}}{4}(r+2)^2 - 3 \cdot \frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{4}(r^2 + 4r + 1).$$

To compute the area of triangle ACE , we can take the area of triangle PQR , and subtract the areas of triangles ACP , CEQ , and AER .



For triangle ACP , if we take $\overline{CP}$ as the base, then the height is the same as the height of equilateral triangle ABP . This height is $\frac{\sqrt{3}}{2}$. Therefore,

$$[ACP] = \frac{1}{2} \cdot (r+1) \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}(r+1).$$

Similarly, $[CEQ] = [AER] = \frac{\sqrt{3}}{4}(r+1)$, so

$$[ACE] = [PQR] - [ACP] - [CEQ] - [AER] = \frac{\sqrt{3}}{4}(r+2)^2 - 3 \cdot \frac{\sqrt{3}}{4}(r+1) = \frac{\sqrt{3}}{4}(r^2 + r + 1).$$

We are told that the area of triangle ACE is 70% of the area of the hexagon $ABCDEF$, so

$$\frac{r^2 + r + 1}{r^2 + 4r + 1} = \frac{7}{10}.$$

This simplifies to $r^2 - 6r + 1 = 0$. The roots are $3 + 2\sqrt{2}$ and $3 - 2\sqrt{2}$. These are both positive, which makes them possible values of r , and their sum is $\boxed{6}$. ■

Exercise. A wire is cut into two pieces, one of length a and the other of length b . The piece of length a is bent to form an equilateral triangle, and the piece of length b is bent to form a regular hexagon. The triangle and the hexagon have equal area. What is $\frac{a}{b}$? (2013 AMC 10B, #15)

Exercise. Each side of a unit square (side length 1) is also one side of an equilateral triangle that lies in the square. Compute the area of the intersection of (the interiors of) all four triangles. (Math Prize for Girls, 2019)

2 3D Geometry

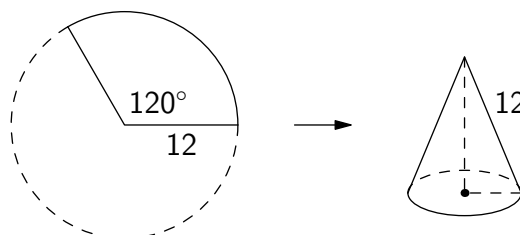
We start by reviewing the formulas for basic three-dimensional geometric figures. These formulas allow us to find the volumes and surface areas of these figures.

Solid	Total Surface Area	Volume
Pyramid		$\frac{1}{3}(\text{base})(\text{height})$
Cylinder	$2\pi rh + 2\pi r^2$	$\pi r^2 h$
Cone	$\pi rl^* + \pi r^2$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$\frac{4}{3}\pi r^3$

(* l is the slant height)

Example. A 120° sector is cut from a circle with a radius of 12, and rolled into a cone. Find the volume of the cone.

Solution. To find the volume of a cone, we need its radius and height.



The circumference of the base of the cone corresponds to the arc of the sector. Since the angle of the sector is 120° , the length of the arc of the sector is $\frac{1}{3}$ the circumference of the whole circle, which is

$$\frac{1}{3} \cdot 2 \cdot \pi \cdot 12 = 8\pi.$$

Hence, the radius of the base of the cone is $\frac{8\pi}{2\pi} = 4$.

The radius of the circle corresponds to the slant height of the cone, which is 12. So by Pythagoras, the height of the cone is

$$\sqrt{12^2 - 4^2} = 8\sqrt{2}.$$

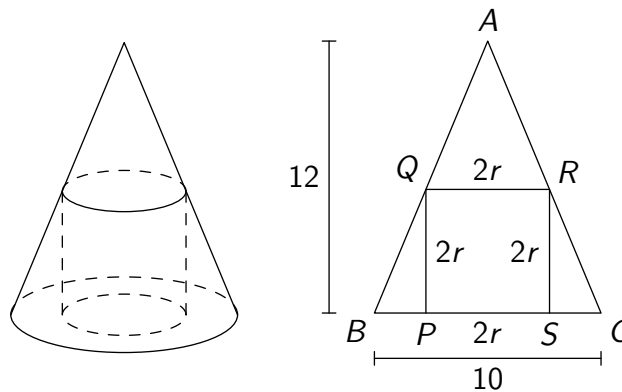
Therefore, the volume of the cone is

$$\frac{1}{3} \pi \cdot 4^2 \cdot 8\sqrt{2} = \boxed{\frac{128\pi\sqrt{2}}{3}}.$$

For many three-dimensional geometry problems, the main technique is to reduce them to two-dimensional geometry problems, by taking an appropriate cross-section.

Example. A right circular cylinder with its diameter equal to its height is inscribed in a right circular cone. The cone has diameter 10 and altitude 12, and the axes of the cylinder and cone coincide. Find the radius of the cylinder. (2001 AMC 10, #21)

Solution. We can take a vertical cross-section.



The cross-section of the cone is an isosceles triangle ABC with base 10 and height 12. Since the height of the cylinder is the same as its diameter, its cross-section is a square $PQRS$. Let the radius of the cylinder be r , so its height is $2r$.

Since $\overline{QR}$ and $\overline{BC}$ are parallel, triangles AQR and ABC are similar. Triangle AQR has a base of $2r$ and a height of $12 - 2r$, so by similarity,

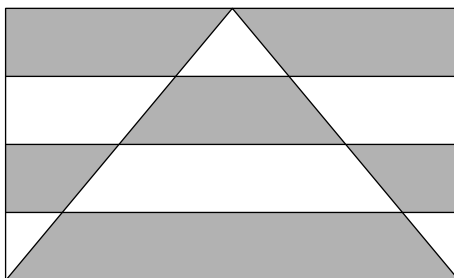
$$\frac{2r}{10} = \frac{12 - 2r}{12}.$$

Solving for r , we find $r = \boxed{\frac{30}{11}}$.

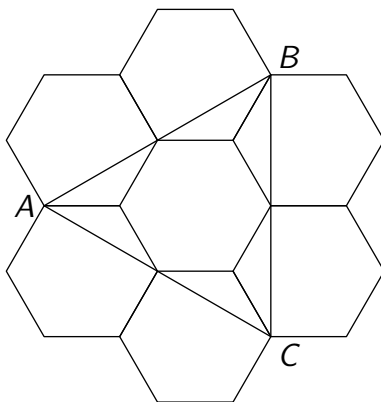
Exercise. A ball was floating in a lake when the lake froze. The ball was removed (without breaking the ice), leaving a hole 24 cm across at the top and 8 cm deep. What was the radius of the ball (in centimeters)? (1987 AHSME, #22)



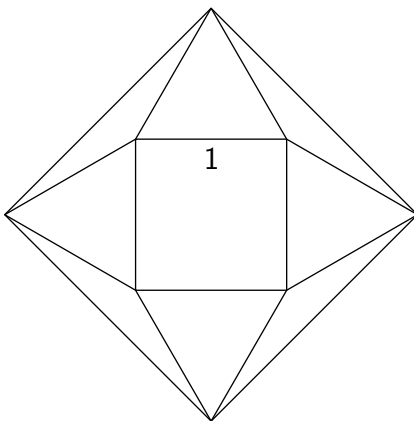
1. The diagram below shows a 12 by 20 rectangle split into four strips of equal widths all surrounding an isosceles triangle. Find the area of the shaded region.



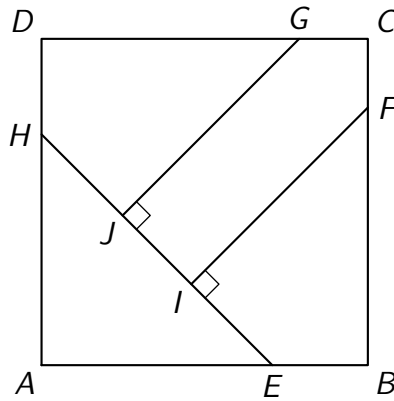
2. Six regular hexagons surround a regular hexagon of side length 1 as shown. What is the area of triangle ABC ?



3. Rhombus $ABCD$ has side length 2 and $\angle B = 120^\circ$. Region R consists of all points inside the rhombus that are closer to vertex B than any of the other three vertices. What is the area of R ?
4. The smaller square in the diagram has side length 1. Equilateral triangles are constructed on its four sides, as shown. What is the area of the larger square?



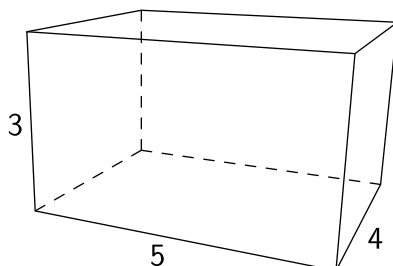
5. Let $ABCDEF$ be a regular hexagon. Let P be the intersection point of $\overline{AC}$ and $\overline{BD}$. Suppose that the area of triangle EPF is 25. What is the area of the hexagon?
6. In square $ABCD$, points E and H lie on $\overline{AB}$ and $\overline{DA}$, respectively, so that $AE = AH$. Points F and G lie on $\overline{BC}$ and $\overline{CD}$, respectively, and points I and J lie on $\overline{EH}$ so that $\overline{FI} \perp \overline{EH}$ and $\overline{GJ} \perp \overline{EH}$. See the figure below. Triangle AEH , quadrilateral $BFIE$, quadrilateral $DHJG$, and pentagon $FCGJI$ each has area 1. What is FI^2 ?



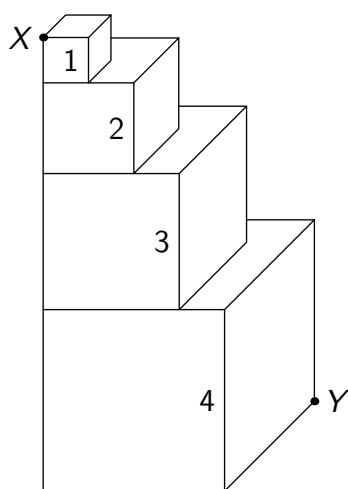
7. Quadrilateral $ABCD$ satisfies $\angle ABC = \angle ACD = 90^\circ$, $AC = 20$, and $CD = 30$. Diagonals $\overline{AC}$ and $\overline{BD}$ intersect at point E , and $AE = 5$. What is the area of quadrilateral $ABCD$?



1. The region consisting of all points in three-dimensional space within 3 units of line segment $\overline{AB}$ has volume 216π . What is the length AB ?
2. The centers of the faces of the right rectangular prism shown below are joined to create an octahedron. What is the volume of this octahedron?

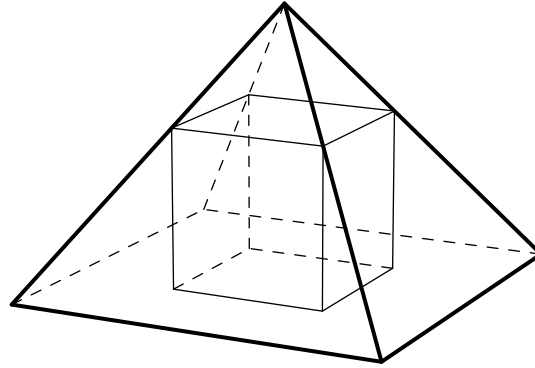


3. Four cubes with edge lengths 1, 2, 3, and 4 are stacked as shown. What is the length of the portion of XY contained in the cube with edge length 3?



4. Nine congruent spheres are packed inside a unit cube in such a way that one of them has its center at the center of the cube and each of the others is tangent to the center sphere and to three faces of the cube. What is the radius of each sphere?

5. A pyramid has a square base with sides of length 1 and has lateral faces that are equilateral triangles. A cube is placed within the pyramid so that one face is on the base of the pyramid and its opposite face has all its edges on the lateral faces of the pyramid. What is the volume of this cube?



6. Let B be a right rectangular prism (box) with edge lengths 1, 3, and 4, together with its interior. For real $r \geq 0$, let $S(r)$ be the set of points in 3-dimensional space that lie within a distance r of some point B . Find the volume of $S(r)$.
7. Tetrahedron $ABCD$ has $AB = 5$, $AC = 3$, $BC = 4$, $BD = 4$, $AD = 3$, and $CD = \frac{12}{5}\sqrt{2}$. What is the volume of the tetrahedron?